

Self-Fulfilling Inflation Risk Premia under the Fiscal Theory: Equilibrium Multiplicity and Government Debt Maturity

Yinjie Yu

September 16, 2026

Abstract

I study self-fulfilling inflation risk premia within the Fiscal Theory of the Price Level. In a New Keynesian model with an affine pricing kernel, the no-arbitrage bond-pricing equation admits *two rational-expectations equilibria* under the same structural parameters, Taylor rule, and fiscal rule: a *Fundamental Equilibrium* in which fiscal uncertainty is unpriced, and a *Fiscal Sunspot Equilibrium* in which investors who coordinate on pricing fiscal uncertainty bid up the real value of government debt. In the Fiscal Sunspot Equilibrium nominal bonds are a fiscal safe haven: they carry a negative fiscal risk premium and a negative inflation risk premium, and a shock to fiscal uncertainty produces an impact deflation followed by a hump-shaped inflation response. Government debt maturity acts asymmetrically on the two channels. Longer duration *dampens* the passthrough of fundamental real-rate shocks to inflation while *amplifying* the self-fulfilling fiscal channel, because the same passthrough factor multiplies the fundamental loading but enters the denominator of the fixed point pinning the fiscal loading. The asymmetry has a direct policy consequence: a maturity shortening of the kind induced by quantitative easing is substantially less disinflationary when fiscal uncertainty is priced than when it is not — roughly half as much at the baseline calibration — and reverses sign outright once the fiscal loading is large enough. The effectiveness of debt maturity management is therefore not a structural constant but depends on which equilibrium investors have coordinated on, with identical fundamentals, policy rules, and states.

1 Introduction

Why do large-scale asset purchases, debt maturity operations, and fiscal-risk episodes have such ambiguous effects on inflation expectations? The same policy appears inflationary in some episodes and disinflationary in others. In some environments government debt behaves like a safe asset, so that a higher debt valuation coincides with *lower* inflation compensation; in others, fiscal news raises inflation expectations and yields together. Standard accounts attribute this to differences in fundamentals — the fiscal position, the policy regime, the state of the business cycle. This paper offers a different explanation: the same fundamentals can support both outcomes, depending on whether investors have coordinated on pricing fiscal risk at all.

The vehicle is the Fiscal Theory of the Price Level (FTPL), where the price level is pinned by the real value of future primary surpluses discounted by equilibrium asset prices. The observation that drives the paper is that the discount factor in that valuation equation is not an external object. If fiscal uncertainty can itself become a priced risk, then beliefs about whether fiscal risk is priced affect bond valuations; bond valuations affect inflation through the government budget constraint; and the resulting inflation dynamics either validate or invalidate the original pricing of fiscal risk. That circularity admits two self-consistent resolutions. Debt duration and quantitative easing then matter not only through the usual term-premium channel, but by changing *which* fiscal risks are capitalized into government debt values.

A closely related puzzle motivates the same mechanism from the asset-pricing side. The inflation risk premium embedded in nominal Treasury yields has been persistently positive for decades, yet standard monetary models tie it to the covariance between nominal bond returns and consumption growth — a covariance that changes sign across regimes while the premium does not.

Under the FTPL, the real value of government debt is determined by the expected present value of future primary surpluses. A non-fundamental shock to perceived fiscal uncertainty — one that need not reflect any change in actual surplus paths — can nonetheless become *self-fulfilling* through a debt revaluation channel. In a New Keynesian model with an affine pricing kernel, I show that the no-arbitrage bond-pricing equation admits two distinct rational-expectations equilibria consistent with the same structural parameters and current fundamentals. In the *Fundamental Equilibrium*, the non-fundamental fiscal uncertainty shock η is entirely unpriced: bond prices load only on the real rate, the stochastic discount factor carries no fiscal risk premium, and η has no effect on inflation, output, or asset prices. In the *Fiscal Sunspot Equilibrium*, η becomes self-fulfilling through a debt revaluation channel: when investors price fiscal uncertainty, the stochastic discount factor loads on η ($\mu \neq 0$), and because the real bond return co-moves *positively* with η ($A_2 > 0$) — the bond gains value precisely when fiscal uncertainty is high and marginal utility is elevated — the bond is a *hedge*. Its fiscal risk premium is therefore *negative* (a safe haven, $RP^\xi < 0$): investors accept

a lower expected return, bidding up the real value of debt. The Fiscal Theory then requires the price level to adjust so that the revalued real debt is consistent with the present value of surpluses — closing the loop. The mechanism is self-validating: the bond is a fiscal hedge only because investors expect it to be priced as one.

Two equilibria, one economy. The coexistence of these equilibria is the paper’s central result. Both are consistent with the same Taylor rule, the same fiscal rule, and the same current values of real rates and debt. They differ only in how investors coordinate on pricing the non-fundamental shock η . In the Fundamental Equilibrium, η is a true sunspot: it drives the fiscal uncertainty state ξ_t but has no equilibrium consequences. In the Fiscal Sunspot Equilibrium, the same shock is self-validating: the real economy responds to η precisely because investors expect it to. The model thus embeds a novel form of indeterminacy — not a multiplicity of price-level *paths*, but a multiplicity of inflation *risk premia* — that arises from the interaction between the FTPL valuation equation and asset-market pricing.

Sunspot and orthogonality. In the Fiscal Sunspot Equilibrium, the η shock is orthogonal to the common macro shock ε that governs real rates and the output gap. This orthogonality is exact by construction: $\text{Cov}(\varepsilon_{t+1}, \eta_{t+1}) = 0$, and in the Fundamental Equilibrium η has no effect on any observable. In the Fiscal Sunspot Equilibrium, η moves the fiscal risk premium and inflation without affecting real rates — a clean separation that the model delivers analytically and that would be difficult to explain in purely monetary frameworks.

Debt maturity acts asymmetrically on the two channels. The second contribution concerns government debt maturity. In the model, the roll-over parameter ω enters the equilibrium through exactly one object: the passthrough factor $PT = 1 - \omega\rho_\pi$, the fraction of a surplus surprise that reaches the price level rather than being absorbed as a capital gain on long bonds. This single factor acts on the two channels in *opposite* directions. In the fundamental channel it multiplies the inflation loading directly, so longer maturity mechanically dampens the passthrough of real-rate shocks to inflation. In the self-fulfilling channel it appears instead in the *denominator* of the fixed point that determines the fiscal loading A_2 , so a lower passthrough sustains a *larger* debt-revaluation loading: longer maturity amplifies the sunspot channel and deepens the safe-haven inflation risk premium. I derive both comparative statics in closed form and verify them against the numerical solver. The asymmetry has a direct policy reading. Maturity shortening of the kind induced by quantitative easing raises the economy’s sensitivity to fundamental real-rate shocks while compressing the self-fulfilling fiscal channel: in the calibration, halving ω cuts the peak inflation response to a fiscal sunspot shock by roughly a half. A central bank that believes the economy has coordinated on the Fiscal Sunspot Equilibrium can therefore use maturity management to dismantle the self-fulfilling mechanism, at the cost of greater exposure to fundamental shocks — a trade-off with no counterpart in the Fundamental Equilibrium.

Optimal maturity is a rule, not a number. The third contribution turns the comparative statics into a policy problem. Taking a second-order approximation to household welfare — which pins the weight on output at $(1+\psi)\kappa/\epsilon_p$ rather than leaving it free — and adding a convenience value for short debt, I ask how maturity should covary with fiscal conditions. Rebalancing at market prices transfers no value, so an anticipated maturity rule works purely by making the passthrough factor state-contingent. Because the fiscal uncertainty shock has conditional volatility $v\sqrt{\xi_t}$, it is large exactly when ξ_t is large, and lengthening maturity in those states absorbs it as a capital gain instead of a price-level move. The optimal slope of duration in fiscal uncertainty is $A_2^2 v^2 P T / (A_1^2 \sigma^2 \rho_\pi)$ — the ratio of the two shocks’ contributions to inflation risk. It is strictly positive when fiscal risk is priced and exactly zero when it is not, so state-contingent maturity policy is valuable *only* in the Fiscal Sunspot Equilibrium. The liquidity value of short debt pulls the other way, because the states in which the government would like to lengthen are the states in which its debt stock, and hence the liquidity it supplies, is largest.

1.1 Related Literature and Contribution

This paper sits at the intersection of three literatures that rarely meet: the Fiscal Theory of the Price Level, self-fulfilling debt crises, and the macro-finance of debt maturity. Its contribution is easiest to state by asking, of each multiplicity result in the literature, *what exactly is indeterminate*. Table 1 organizes the comparison.

Table 1: What is indeterminate? Positioning relative to the multiplicity literature.

Literature	What is indeterminate	Sign of the risk premium	Role of debt maturity
Self-fulfilling debt crises [Calvo, 1988, Cole and Kehoe, 2000, Aguiar et al., 2022]	Default / rollover outcome, via the solvency constraint	Positive (crisis)	Secondary
Debt dilution [Aguiar and Amador, 2020]	Debt accumulation path, via beliefs about future creditor pricing	Positive	<i>Necessary:</i> multiplicity requires long bonds
Safe-asset status [Brunnermeier et al., 2022, 2020]	Whether debt carries a bubble / service-flow premium	Positive (vicious loop)	Secondary
Monetary indeterminacy [Benhabib et al., 2001, Davig and Leeper, 2006]	Steady state or policy regime	Not priced	Absent
This paper	Whether fiscal uncertainty is priced in the SDF at all	Negative (hedge)	Amplifies, but is not necessary

Multiplicity in what is priced, not in prices. The multiplicity here is not in the price level,

not in the default decision, not in a bubble term, and not in the path of debt. Both equilibria feature the same fundamentals, the same fiscal and monetary rules, the same state variables, and a well-defined finite present value of surpluses. They differ in a single object: the loading μ of the pricing kernel on a non-fundamental shock. To our knowledge this is the first result in which the indeterminacy is over *whether a given risk commands a price*, rather than over the level or path of any price. That framing is what makes the two equilibria observationally distinguishable — they differ in risk premia, not in average inflation.

The sign is the opposite of the crisis literature. In Calvo [1988], Cole and Kehoe [2000] and Brunnermeier et al. [2022], the self-fulfilling loop runs through a *positive* premium: investors believe debt is risky, demand compensation, raise the fiscal burden, and thereby make debt riskier. The loop in this paper runs the other way. Because the bond gains real value exactly when fiscal uncertainty is high, it is a *hedge*, its fiscal risk premium is *negative*, and the self-validating equilibrium is the one in which government debt is *more* valuable, not less. This is the mirror image of the vicious circle, and it is why the Fiscal Sunspot Equilibrium produces an impact *deflation* rather than a fiscal-stress inflation. Section 3.4 makes the point explicit, since readers who arrive from the crisis literature will expect the opposite sign.

Maturity amplifies the loop but does not create it. Aguiar and Amador [2020] show that with long-maturity bonds, creditor beliefs about future pricing become self-fulfilling, and long maturity is *necessary* for their multiplicity: at one-period maturity the indeterminacy disappears. The mechanism here is different. Setting $\omega = 0$ leaves both equilibria intact ($A_2^* = 0.142 > 0$), so the multiplicity does not rest on duration. What duration does is *amplify* the self-fulfilling channel while simultaneously *dampening* the fundamental one (Section 5.3) — an asymmetry that has no analogue in the dilution mechanism.

FTPL with risk premia, but with a unique equilibrium. The foundational FTPL papers [Sims, 1994, Woodford, 1994, Leeper, 1991] establish that the price level can be determined by the government’s intertemporal budget constraint, and Cochrane [2001, 2023] develop the modern treatment. A recent literature embeds that valuation equation in an asset-pricing framework: Jiang et al. [2023] and Jiang et al. [2024] show that discounting surpluses at risk-adjusted rates substantially changes measured fiscal capacity, and Croce et al. [2012], Kung [2015] price fiscal uncertainty in the term structure. Bianchi et al. [2023] show empirically that unfunded fiscal shocks drive persistent inflation. All of these maintain a *unique* equilibrium in which fiscal risk is priced by fundamentals. The present paper does not dispute the pricing; it asks whether the pricing is pinned down at all, and shows it is not.

Debt maturity and the transmission of QE. Corhay et al. [2023] is the closest antecedent for the second contribution: they show in a unique-equilibrium macro-finance model that maturity operations move expected inflation through the conditional nominal term premium. Empirically,

Krishnamurthy and Vissing-Jorgensen [2011], Du et al. [2024] and Haddad et al. [2025] document the channels and real effects of asset purchases, and Burlon et al. [2025] studies the exit. Relative to Corhay et al. [2023], this paper adds two things. First, the term-premium channel is quantitatively small here, while the dominant effect runs through the repricing of the government balance sheet via the FTPL constant condition (Table 6). Second, and more importantly, the *magnitude* of the inflation response to a maturity operation depends on which equilibrium investors have coordinated on — the same shock is roughly half as disinflationary in the Fiscal Sunspot Equilibrium — and the effect reverses sign once the fiscal loading is large enough. A unique-equilibrium framework cannot generate that state dependence, which is precisely the empirical puzzle that motivates the paper.

Inflation risk premia. The term-structure literature documents that the inflation risk premium is persistently positive and more stable than the nominal term premium [Haubrich et al., 2012, Campbell et al., 2013], and that the nominal-real covariance switches sign across regimes while the premium does not. The two equilibria here offer a candidate account: the Fundamental Equilibrium supplies a fundamentals-driven lower bound, while the Fiscal Sunspot Equilibrium adds a self-validating component driven by a non-fundamental shock. Under that reading the observed premium is not uniquely pinned by fundamentals, which is consistent with its imperfect co-movement with measured macroeconomic risk. Khorrami and Mendo [2024] share the self-fulfilling structure in a different setting, where the loop runs through balance-sheet amplification rather than the government budget constraint.

Paper organization. Section 2 develops the model and derives the two equilibria and the self-fulfilling mechanism. Section 4 reports the calibration and the implied equilibrium coefficients. Section 5 characterizes the impulse responses to each shock and establishes the asymmetric maturity result in Section 5.3. Section 6 sets up the government’s maturity problem and derives an optimal state-contingent maturity rule, contrasting an anticipated rule with an unanticipated one. Section 7 discusses equilibrium selection and Section 8 the testable implications. Section 9 concludes. Technical derivations are in the Appendix.

2 The Model

This section states the environment and the problem each agent solves. Nothing here presumes an equilibrium; the equilibrium objects are characterized in Section 3.

The economy has four types of agent: a representative household that consumes, supplies labor, and holds government debt; a continuum of monopolistically competitive firms with Calvo price setting; a government that issues long-maturity nominal debt and sets the primary surplus by rule; and a central bank that sets the nominal interest rate by rule.

2.1 Environment

Time is discrete, $t = 0, 1, 2, \dots$. Two orthogonal unit-normal shocks drive all aggregate risk,

$$\varepsilon_{t+1} \sim \mathcal{N}(0, 1), \quad \eta_{t+1} \sim \mathcal{N}(0, 1), \quad \text{Cov}(\varepsilon_{t+1}, \eta_{t+1}) = 0,$$

and two exogenous state variables summarize the aggregate state.

Real rate. The real short rate follows a Gaussian AR(1),

$$r_{t+1} = \bar{r} + \rho_r(r_t - \bar{r}) + \sigma \varepsilon_{t+1}, \quad \rho_r \in (0, 1), \sigma > 0. \quad (1)$$

Fiscal uncertainty. The second state $\xi_t \geq 0$ follows a square-root (CIR-type) process,

$$\xi_{t+1} = \bar{\xi} + \rho_\xi(\xi_t - \bar{\xi}) + \nu \sqrt{\xi_t} \eta_{t+1}, \quad \bar{\xi} > 0, \rho_\xi \in (0, 1), \nu > 0. \quad (2)$$

All conditional moments of ξ_{t+1} are linear in ξ_t , which preserves the affine structure used in Section 3.

What ξ_t is, and is not. It is worth being explicit at the outset, because ξ_t is the object the paper's multiplicity is about. ξ_t enters *no* primitive of the model: not technology, not preferences, not the surplus rule, not either policy rule. It appears nowhere in Sections 2.2–2.5. The realized path of surpluses, output and technology is therefore invariant to η , and in an equilibrium where ξ_t is unpriced the shock has no observable consequence whatsoever. The distinction that matters is between

1. **fundamental fiscal risk** — actual uncertainty about future primary surpluses, which this model deliberately does not contain; and
2. **coordinated fiscal-risk pricing** — whether agents' stochastic discount factor loads on ξ_t .

The multiplicity established below is entirely about the second. Empirically ξ_t is best read as a *coordination state*: a publicly observed variable — a narrative, a salient fiscal episode, a market

convention — that investors may or may not treat as priced. Section 7 returns to this.

2.2 Households

The representative household chooses consumption C_t , labor L_t , nominal holdings of government debt B_t^h , and a real risk-free asset a_t in zero net supply, to solve

$$\max_{\{C_t, L_t, B_t^h, a_t\}} \mathbb{E}_0 \sum_{t \geq 0} \beta^t \left(\log C_t - \frac{L_t^{1+\psi}}{1+\psi} \right), \quad \beta \in (0, 1), \psi > 0, \quad (3)$$

subject to the sequence of nominal budget constraints

$$P_t C_t + Q_t B_t^h + P_t a_t + P_t s_t = W_t L_t + \Pi_t + [(1 - \omega) + \omega Q_t] B_{t-1}^h + P_t a_{t-1} (1 + r_{t-1}), \quad (4)$$

together with a no-Ponzi condition. Here P_t is the price level, W_t the nominal wage, Π_t lump-sum profits rebated by firms, s_t the real primary surplus levied as a lump-sum tax, and Q_t the price of the government bond described in Section 2.4. The bracketed term is the payoff on debt carried into t : a coupon $(1 - \omega)$ plus the market value ωQ_t of the portion that rolls over.

Optimality. Let $M_{t,t+1} \equiv \beta C_t / C_{t+1}$ denote the real stochastic discount factor and $m_{t+1} \equiv \log M_{t,t+1}$. The intertemporal conditions require that M price every traded asset,

$$\mathbb{E}_t [M_{t,t+1} R_{t+1}^x] = 1 \iff \log \mathbb{E}_t [\exp(m_{t+1} + r_{t+1}^x)] = 0 \quad (5)$$

for the gross real return R_{t+1}^x on each of the bond and the risk-free asset, and the intratemporal condition gives the labor supply schedule

$$\frac{W_t}{P_t} = C_t L_t^\psi. \quad (6)$$

Log utility is not innocuous here and is worth flagging as a modelling choice: combined with goods-market clearing it makes the SDF innovation exactly the negative of the output innovation,

$$m_{t+1} - \mathbb{E}_t[m_{t+1}] = -(y_{t+1} - \mathbb{E}_t[y_{t+1}]), \quad y_t \equiv \log Y_t, \quad (7)$$

which is the channel through which inflation dynamics feed back into the pricing kernel.

2.3 Firms

A unit continuum of firms indexed by i produces differentiated varieties aggregated into final output by a CES aggregator with elasticity $\epsilon_p > 1$, so that firm i faces

$$Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\epsilon_p} Y_t, \quad P_t = \left[\int_0^1 P_t(i)^{1-\epsilon_p} di \right]^{\frac{1}{1-\epsilon_p}}. \quad (8)$$

Production is linear in labor, $Y_t(i) = A_t L_t(i)$, so real marginal cost $mc_t = W_t/(A_t P_t)$ is common across firms.

Calvo price setting. Each period a randomly drawn fraction $1 - \theta$ of firms may reset its price; the rest keep last period's. A firm resetting at t chooses P_t^* to maximize the present discounted value of real profits over the random period for which the price applies,

$$\max_{P_t^*} \mathbb{E}_t \sum_{k=0}^{\infty} \theta^k M_{t,t+k} \left[\frac{P_t^*}{P_{t+k}} - mc_{t+k} \right] Y_{t+k|t}, \quad Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\epsilon_p} Y_{t+k}, \quad (9)$$

where $M_{t,t+k} = \prod_{j=1}^k M_{t+j-1,t+j}$ is the household's multi-period SDF. Firms are owned by the household, so *they discount with the same kernel that prices government debt*. This is the standard assumption, but in this model it has bite: any risk the kernel prices also shows up in firms' pricing decisions, and Section 3 shows it leaves a residual term in the Phillips curve.

Optimality. The first-order condition for P_t^* is

$$\mathbb{E}_t \sum_{k=0}^{\infty} \theta^k M_{t,t+k} Y_{t+k|t} \left[\frac{P_t^*}{P_{t+k}} - \mathcal{M} mc_{t+k} \right] = 0, \quad \mathcal{M} \equiv \frac{\epsilon_p}{\epsilon_p - 1}, \quad (10)$$

which is conveniently written with two auxiliary recursions,

$$\frac{P_t^*}{P_t} = \mathcal{M} \frac{\mathcal{N}_t}{\mathcal{D}_t}, \quad \begin{aligned} \mathcal{N}_t &= Y_t mc_t + \theta \mathbb{E}_t \left[M_{t,t+1} \Pi_{t+1}^{\epsilon_p} \mathcal{N}_{t+1} \right], \\ \mathcal{D}_t &= Y_t + \theta \mathbb{E}_t \left[M_{t,t+1} \Pi_{t+1}^{\epsilon_p-1} \mathcal{D}_{t+1} \right], \end{aligned} \quad (11)$$

with $\Pi_{t+1} \equiv P_{t+1}/P_t$. The aggregate price index evolves as $P_t^{1-\epsilon_p} = \theta P_{t-1}^{1-\epsilon_p} + (1-\theta)(P_t^*)^{1-\epsilon_p}$. The two recursions differ only by one power of gross inflation, which is exactly why their risk corrections fail to cancel.

2.4 Government and Debt

The government issues a single *geometric-coupon* nominal bond indexed by $\omega \in [0, 1)$. One unit of face value outstanding pays a coupon $(1 - \omega)$ next period and leaves ω units outstanding, so a unit

issued at t delivers the declining stream $(1 - \omega), (1 - \omega)\omega, (1 - \omega)\omega^2, \dots$. This single parameter indexes the whole maturity structure: $\omega = 0$ is one-period debt and $\omega \rightarrow 1$ a consol. Writing B_t for face value outstanding and Q_t for the real price, the real market value of debt is $b_t \equiv Q_t B_t$ and the real gross return is

$$r_{g,t+1} = \frac{(1 - \omega) + \omega Q_{t+1}}{Q_t}. \quad (12)$$

At $\omega = 0$ this is $1/Q_t$, known at t : a one-period bond has no capital-gain component. For $\omega > 0$ the resale value ωQ_{t+1} makes the return risky.

Budget constraint. The government's *flow* budget constraint — not a balanced-budget requirement, since deficits may run indefinitely — is, in real terms,

$$\frac{b_{t+1} + s_t}{b_t} = \frac{r_{g,t+1}}{1 + \pi_{t+1}}, \quad (13)$$

where π_{t+1} is inflation between t and $t + 1$.

Fiscal rule. The primary surplus responds to outstanding debt with elasticity δ_b ,

$$s_{t+1} = s^* \left(\frac{b_t}{b^*} \right)^{\delta_b}, \quad s^*, b^* > 0, \quad \delta_b \in (0, 1). \quad (14)$$

Policy is *active* in the sense of [Leeper \[1991\]](#): with $\delta_b < 1$ the surplus does not fully stabilize debt, so the intertemporal budget constraint is an equilibrium condition on the price level rather than a constraint the Treasury meets by adjusting taxes. Log-linearizing (13) with (14) gives the log real surplus return (Appendix A.1)

$$r_{s,t+1} = k_0 + k_1 \log b_{t+1} + (k_2 \delta_b - 1) \log b_t, \quad k_1 = \frac{b^*}{b^* + s^*}, \quad k_2 = \frac{s^*}{b^* + s^*}, \quad (15)$$

with $k_1 + k_2 = 1$ and k_0 a constant fixed by (b^*, s^*) alone.

Timing. Table 2 fixes the convention, which matters for reading the long-bond equations.

Table 2: Timing convention.

Object	Determined at	Meaning
B_t	end of t	Face value outstanding entering $t + 1$; predetermined at $t + 1$
Q_t	t	Ex-coupon real price of the geometric bond
$b_t = Q_t B_t$	t	Real market value of debt, $\mathcal{F}_t$ -measurable
s_t	t	Real primary surplus in period t
$r_{g,t+1}$	$t + 1$	Real return on debt held from t to $t + 1$, eq. (12)
π_{t+1}	$t + 1$	Inflation between t and $t + 1$
i_t	t	Nominal rate set at t , paid on debt serviced at $t + 1$

All debt is held by the household; there is no separate central-bank balance sheet and no

reserves. Consequently ω is the maturity of the *consolidated* government's liabilities, and a change in ω should be read as an operation of the consolidated government rather than as an asset swap between the Treasury and a central bank. Section 5.3 discusses what this does and does not license.

2.5 Monetary Policy

The central bank sets the nominal short rate by a Taylor rule,

$$i_t = i^* + \rho_\pi(\pi_t - \pi^*), \quad \rho_\pi \in (0, 1). \quad (16)$$

Policy is *passive*: the nominal rate responds to inflation less than one for one, so it does not raise the real rate enough to pin the price level on its own. That role falls to the government's intertemporal budget constraint. Together with active fiscal policy this is the Leeper regime in which the Fiscal Theory operates. Imposing $\rho_\pi > 1$ would both overdetermine the system and, as Section 5.5 shows, render the equilibrium inflation recursion explosive.

2.6 Market Clearing and Equilibrium

Goods, labor, bond and risk-free asset markets clear:

$$C_t = Y_t, \quad L_t = \int_0^1 L_t(i) di, \quad B_t^h = B_t, \quad a_t = 0. \quad (17)$$

Definition 1. A *stationary rational-expectations equilibrium* is a collection of stochastic processes $\{C_t, L_t, Y_t, P_t, \pi_t, i_t, Q_t, B_t, b_t, s_t, m_{t+1}\}$, adapted to the filtration generated by (r_t, ξ_t) , such that

1. the household's conditions (54)–(6) hold given prices;
2. resetting firms solve (56) and the price index aggregates (11);
3. the government satisfies (13) with surpluses given by (14), and the central bank follows (16);
4. markets clear as in (17); and
5. all processes are stationary.

The analysis below restricts attention to equilibria in which the log real value of debt is *affine* in the two state variables. This is a genuine restriction and Section 3.4 is explicit about what it does and does not deliver.

3 Equilibrium Characterization

3.1 Duration and the Passthrough Factor

Maturity enters the equilibrium through exactly one object, and isolating it first makes everything that follows a one-period coefficient rescaled by that object.

An inflation surprise at $t + 1$ raises the nominal rate through the Taylor rule (16), depressing the resale price of the long bond and converting part of the surprise into a capital loss. Because inflation itself mean-reverts at ρ_π , the induced rate path decays geometrically, and the bond capitalizes it with weights ω^k (Appendix A.3):

$$r_{g,t+1} - \mathbb{E}_t[r_{g,t+1}] = -\gamma(\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]), \quad \gamma \equiv \frac{\omega\rho_\pi}{1 - \omega\rho_\pi}. \quad (18)$$

Substituting into the unexpected component of (13),

$$\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}] = -PT(r_{s,t+1} - \mathbb{E}_t[r_{s,t+1}]), \quad PT \equiv \frac{1}{1 + \gamma} = 1 - \omega\rho_\pi. \quad (19)$$

The *passthrough factor* PT is the share of a surplus surprise that reaches the price level rather than being absorbed as a capital gain.

Duration is bounded. Because $\rho_\pi < 1$, letting $\omega \rightarrow 1$ gives $\omega\rho_\pi \rightarrow \rho_\pi$, not 1. Hence

$$\gamma \leq \frac{\rho_\pi}{1 - \rho_\pi}, \quad PT \geq 1 - \rho_\pi > 0. \quad (20)$$

Even a consol has finite inflation-feedback duration, because the rate response it capitalizes is itself transitory. Unbounded γ would require $\rho_\pi \rightarrow 1$, which is precisely the boundary at which monetary policy ceases to be passive. This has an economic reading worth stating: *under passive monetary policy there is a floor $1 - \rho_\pi$ on the passthrough of fiscal shocks to inflation, whatever the maturity structure.* The price level cannot be insulated from fiscal news by debt management alone.

3.2 Affine Solution and Coefficient Conditions

Conjecture that the log real value of debt is affine,

$$\log b_t = A_0 + A_1 r_t + A_2 \xi_t, \quad (21)$$

and that the log SDF is affine in the shocks,

$$m_{t+1} = -\delta - r_t - \lambda \varepsilon_{t+1} - \mu \sqrt{\xi_t} \eta_{t+1}, \quad (22)$$

with δ the Jensen term ensuring $\log \mathbb{E}_t[\exp(m_{t+1})] = -r_t$. The loadings λ, μ are the market prices of real-rate and fiscal risk; they are *not* free parameters and are pinned in Section 3.3. Substituting (21) and (22) into the bond Euler equation (54) and matching on r_t, ξ_t and the constant gives (Appendix A.4)

$$(r_t): \quad A_1(k_1\rho_r + k_2\delta_b - 1) = 1 \quad (23)$$

$$(\xi_t): \quad A_2 \underbrace{\left(k_1\rho_\xi + k_2\delta_b - 1 + \frac{1}{2}k_1^2 A_2 v^2 - \mu k_1 v\right)}_{=-N} = 0 \quad (24)$$

$$\begin{aligned} (\text{const}): \quad & k_0 + (k_1 + k_2\delta_b - 1)A_0 + \frac{1}{2}(k_1 A_1 \sigma)^2 - \lambda k_1 A_1 \sigma \\ & + k_1 A_1 (1 - \rho_r)\bar{r} + k_1 A_2 (1 - \rho_\xi)\bar{\xi} = 0 \end{aligned} \quad (25)$$

where $N \equiv 1 - k_1\rho_\xi - k_2\delta_b$, assumed positive. Condition (23) gives immediately

$$A_1 = \frac{1}{k_1\rho_r + k_2\delta_b - 1} < 0, \quad (26)$$

negative by the active-fiscal stability condition $k_1\rho_r + k_2\delta_b < 1$, and independent of ω . Higher real rates depress the present value of surpluses and hence the market value of debt. Condition (24) is the heart of the paper: it has the root $A_2 = 0$, and a second branch obtained by setting the bracket to zero,

$$\mu = \frac{k_1 v}{2} A_2 - \frac{N}{k_1 v}, \quad (27)$$

which expresses the fiscal risk price as a *linear* function of the loading A_2 . Whether that branch is consistent with the rest of the model depends on what pins μ independently — which is the household and firm side, taken up next.

3.3 Risk Prices and the Phillips Curve

Log-linearizing the Calvo block (11) around a zero-inflation steady state and combining with the price index gives the Phillips curve (Appendix A.2),

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa \widehat{mc}_t + \Lambda_t, \quad \kappa \equiv \frac{(1 - \theta)(1 - \beta\theta)}{\theta}, \quad (28)$$

where $\widehat{mc}_t$ is the log deviation of real marginal cost. Two features deserve emphasis.

θ is not free. The slope κ and the reset probability are tied by $\kappa = (1 - \theta)(1 - \beta\theta)/\theta$, so choosing κ pins θ as the root in $(0, 1)$ of $\beta\theta^2 - (1 + \beta + \kappa)\theta + 1 = 0$. At the calibration below $\kappa = 0.10$ implies $\theta = 0.733$, a mean price duration of 3.7 quarters.

The risk wedge Λ_t . In a first-order approximation Λ_t is identically zero: the stochastic part of

$M_{t,t+k}$ multiplies terms that are already first-order, so it drops by certainty equivalence. But this model retains second-order terms elsewhere — they are what generate the second root of (24) — and consistency requires retaining them here too. Because $\mathcal{N}_t$ and $\mathcal{D}_t$ in (11) carry different powers of inflation, their Jensen corrections do not cancel, leaving

$$\Lambda_t = \beta(1 - \theta) \left[\frac{2\varepsilon_p - 1}{2} \text{var}_t(\pi_{t+1}) + \text{cov}_t(m_{t+1}, \pi_{t+1}) \right]. \quad (29)$$

Both terms are affine in ξ_t , so $\Lambda_t = \Lambda_0 + \Lambda_\xi \xi_t$. The wedge is a direct consequence of firms discounting with the households' kernel: if fiscal risk is priced, firms' reset decisions inherit it.

Risk prices. Writing the model's demand block as $\kappa \widehat{m}_t = \kappa(y_t - y_n) + \psi_r r_t$ and inverting (28),

$$y_t = y_n - \frac{\psi_r}{\kappa} r_t + \frac{1}{\kappa} \left(\pi_t - \beta \mathbb{E}_t[\pi_{t+1}] - \Lambda_t \right). \quad (30)$$

Differencing forward and using the equilibrium recursion (39), the innovation in $\mathbb{E}_{t+1}[\pi_{t+2}]$ has *three* components,

$$\mathbb{E}_{t+1}[\pi_{t+2}] - \mathbb{E}_t[\mathbb{E}_{t+1}[\pi_{t+2}]] = \rho_\pi \Delta \pi_{t+1} \underbrace{- \sigma \varepsilon_{t+1}}_{\text{from } -r_{t+1}} + \underbrace{(B_3 + B_4 \rho_\xi) v \sqrt{\xi_t} \eta_{t+1}}_{\text{from } \xi_{t+1}}, \quad (31)$$

writing $\Delta \pi_{t+1} \equiv \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$. Retaining only the first understates both risk prices. Substituting into (30) and imposing $m_{t+1} - \mathbb{E}_t[m_{t+1}] = -(y_{t+1} - \mathbb{E}_t[y_{t+1}])$ from (55) gives (Appendix A.5)

$$\boxed{\lambda = \frac{\sigma}{\kappa} [(1 - \beta \rho_\pi) B_2 + \beta - \psi_r], \quad \mu = \frac{v}{\kappa} [(1 - \beta \rho_\pi) B_4 - \beta (B_3 + B_4 \rho_\xi) - \Lambda_\xi]} \quad (32)$$

This is the second, independent determination of μ . It is a pure equilibrium object: it invokes the Phillips curve, the Taylor-rule persistence, and the equilibrium inflation loadings, none of which appears in the SDF by assumption.

3.4 The Two Equilibria

Equations (27) and (32) are two expressions for the same object μ , one from the bond market and one from the goods market. Equilibrium requires them to agree.

Why uniqueness fails. Leeper [1991] and Cochrane [2001] establish that the active-fiscal, passive-monetary regime pins the price level uniquely through $B_t/P_t = \mathbb{E}_t \sum_{j \geq 0} m_{t,t+j} s_{t+j}$, because B_t is predetermined and the right-hand side is tied to fiscal fundamentals. The hidden premise is that $m_{t,t+j}$ is itself determined by fundamentals. It is not: the kernel must price every risk bondholders bear, so whether inflation responds to η depends on whether debt values load on ξ_t , which depends on whether the kernel prices ξ_t , which depends on whether inflation responds to η . That circularity

is what (27) and (32) formalize, and it has more than one resolution.

Reduction to a quadratic. Substituting (27) into (32), with $B_4 = -PT k_1 A_2$ and $B_3 + B_4 \rho_\xi$ from (38), every term is at most quadratic in A_2 (Appendix A.6). The non-zero equilibria therefore solve

$$\mathcal{A} A_2^2 + \mathcal{B} A_2 - c_0 = 0, \quad c_0 \equiv \frac{N}{k_1 v} > 0, \quad (33)$$

with $\mathcal{A} = (\beta k_1^2 v^3 PT / \kappa) S$ and

$$S = \frac{\gamma + 1 - \theta}{2} - \frac{\gamma}{1 + \gamma} - \frac{\gamma^2}{2(1 + \gamma)} + \frac{(1 - \theta)(2\epsilon_p - 1)}{2(1 + \gamma)}. \quad (34)$$

Proposition 1. *Assume passive monetary policy ($\rho_\pi < 1$), active fiscal policy ($\delta_b < 1$), $k_1 \rho_r + k_2 \delta_b < 1$ and $N > 0$. Within the affine class (21)–(22):*

1. $A_2 = 0, \mu = 0$ is always an equilibrium — the Fundamental Equilibrium.
2. Because the constant term of (33) is $-c_0 < 0$, the product of its roots is $-c_0/\mathcal{A}$. Hence if $S > 0$ there is exactly one strictly positive root, and if $S < 0$ there are two.
3. Any positive root has $\mu < 0$: the Fiscal Sunspot Equilibrium, in which government debt is a fiscal hedge.

The economically relevant case is the smallest positive root. When $S < 0$ the second positive root occurs at an A_2 two orders of magnitude larger, where the conditional variance of inflation is so large that the log-normal approximation underlying the Jensen terms has long since failed; Section 5.5 quantifies this and Appendix A.6 gives the argument for discarding it. With that restriction the model has exactly two equilibria.

Two equilibria, or two affine equilibria? Proposition 1 is a statement *within* the affine Markov class conjectured in (21). It does not rule out non-affine or non-Markov equilibria, and (24) is silent about them. The substantive claim is not that exactly two equilibria exist globally, but that *even within the most disciplined solution class, whether fiscal uncertainty is priced is not pinned down by fundamentals*.

The self-fulfilling loop. The Fiscal Sunspot Equilibrium is sustained by three links:

1. **Belief $\rightarrow$ debt value.** If the kernel prices fiscal uncertainty ($\mu < 0$), debt values rise with ξ_t : $A_2 > 0$ in (21).
2. **Debt value $\rightarrow$ inflation.** The budget constraint requires b_t to equal the present value of surpluses. If b_t rises with surpluses unchanged, the price level falls on impact ($B_4 < 0$), and the elevated premium then propagates forward through the passive Taylor rule ($B_3 + B_4 \rho_\xi > 0$).

3. **Inflation** $\rightarrow$ **belief**. If inflation responds to η , bondholders face genuine inflation risk. Log utility and the Phillips curve force the kernel to price it, validating $\mu < 0$.

In the Fundamental Equilibrium each link closes at zero.

This is not a default channel. The negative risk premium runs against the intuition from the sovereign-debt literature, where fiscal risk means higher yields and a *positive* premium. That intuition applies to a channel this model does not contain: there is no default, no solvency constraint and no rollover risk, and the only adjustment margin is the price level. What is priced is the state in which FTPL revaluation pushes the price level *down* — a low-inflation, low-output, high-marginal-utility state. A nominal bond that gains real value there is insurance, so investors accept a lower expected return. The fiscal risk premium

$$\text{RP}_t^\xi \equiv -\text{Cov}_t(m_{t+1}, r_{s,t+1})|_\eta = \mu k_1 A_2 v \xi_t < 0 \quad (35)$$

is the mirror image of the vicious circle in Calvo [1988], Cole and Kehoe [2000] and Brunnermeier et al. [2022], where believing debt risky raises the burden and validates the belief. Here, believing fiscal risk priced makes debt *more* valuable and validates that. A model with default, or with surpluses falling in the bad state, would deliver $\text{RP}^\xi > 0$ and reverse the loop.

3.5 Inflation Dynamics

Substituting the affine debt value (21) into the surplus return (15) and using the passthrough relation (19) delivers the inflation innovation directly; matching coefficients in the budget identity (Appendix A.3) gives the exact affine recursion

$$\pi_{t+1} = \rho_\pi \pi_t + B_0 + B_1 r_t + B_2 r_{t+1} + B_3 \xi_t + B_4 \xi_{t+1}, \quad (36)$$

with

$$B_2 = -k_1 A_1 P T > 0, \quad B_4 = -k_1 A_2 P T \leq 0, \quad B_1 = -1 - B_2 \rho_r < 0, \quad (37)$$

and, collecting the ξ_t terms of the bond-Euler equation with $s \equiv B_4 v$,

$$B_3 + B_4 \rho_\xi = \underbrace{A_2 N}_{\text{FTPL fiscal channel}} - \underbrace{\gamma s(\mu + s) - \frac{1}{2} \gamma^2 s^2}_{\text{Jensen correction}}. \quad (38)$$

The Fisher condition $B_1 + B_2 \rho_r = -1$ holds exactly for every ω , and the Jensen correction vanishes at $\omega = 0$. Taking conditional expectations,

$$\mathbb{E}_t[\pi_{t+1}] = \rho_\pi \pi_t + C_0 - r_t + (B_3 + B_4 \rho_\xi) \xi_t, \quad (39)$$

with $C_0 = B_0 + B_2(1 - \rho_r)\bar{r} + B_4(1 - \rho_\xi)\bar{\xi}$. Three channels drive inflation: a *backward* channel $(\rho_\pi\pi_t, B_1r_t, B_3\xi_t)$ reflecting the nominal interest cost of outstanding debt; an *FTPL revaluation* channel $(B_2r_{t+1}, B_4\xi_{t+1})$ in which unexpected changes in debt values force immediate price-level adjustment, both damped by PT ; and the *fiscal sunspot* channel $(B_3 + B_4\rho_\xi > 0)$, present only when $A_2 > 0$.

Inflation risk premium. Define $\text{IRP}_t \equiv i_t - \mathbb{E}_t[\pi_{t+1}] - r_t$. Substituting the Taylor rule and (39),

$$\text{IRP}_t = -(B_3 + B_4\rho_\xi)\xi_t, \quad (40)$$

identically zero in the Fundamental Equilibrium and strictly negative in the Fiscal Sunspot Equilibrium: nominal bonds are a safe haven against fiscal uncertainty, and investors accept a yield concession to hold them.

4 Calibration

The model operates in the *active fiscal / passive monetary* regime required by the Fiscal Theory: monetary policy is passive ($\rho_\pi < 1$), leaving price-level determination to the government's intertemporal budget constraint, while the fiscal rule is active ($\delta_b < 1$), so the surplus does not fully offset debt deviations. Imposing $\rho_\pi > 1$ instead would pin the price level through monetary policy alone, making the fiscal no-arbitrage condition redundant and rendering the inflation recursion (36) explosive.

Table 3 reports the baseline parameter set; all parameters are quarterly. The real-rate process follows standard estimates for the short-term real rate. Fiscal parameters target a debt-to-GDP ratio of 60% with a steady-state surplus satisfying $s^*/b^* = \bar{r}$, so that the steady-state real return on debt is internally consistent. The fiscal-uncertainty block $(\nu, \bar{\xi}, \rho_\xi)$ is set so that the η shock has roughly the same impact on inflation as the ε shock, making the two channels directly comparable.

Table 4 reports the equilibrium objects implied by these parameters in both equilibria. All entries are generated directly by the solver (`src/scripts/make_paper_tables.py`), so the table cannot drift from the code.

A caveat on the fiscal-uncertainty block. The parameters $(\nu, \bar{\xi}, \rho_\xi)$ are the least well identified in the model, and they are being asked to do three things at once: set the size of the η shock relative to the ε shock, deliver a plausible inflation risk premium, and keep the CIR process positive. They cannot do all three. At the values in Table 3 the unconditional standard deviation of ξ_t is $\nu\sqrt{\bar{\xi}/(1 - \rho_\xi^2)}$, which is roughly three times $\bar{\xi}$ itself, so the Feller-type positivity requirement is violated by a wide margin and $\sqrt{\xi_t}$ is not well defined over part of the state space. Lowering ν restores positivity but inflates A_2^* and the implied risk premium; raising it does the reverse. Nothing in Sections 2–3.4 depends on the resolution — the multiplicity and all comparative-

Table 3: Baseline Calibration

Parameter	Symbol	Value	Rationale
<i>Real rate process</i>			
Steady-state real rate	$\bar{r}$	0.010	1% quarterly ($\approx 4\%$ annual)
Persistence	ρ_r	0.90	Standard quarterly estimate
Innovation volatility	σ	0.005	≈ 50 bp annual s.d.
<i>Fiscal uncertainty (CIR)</i>			
Steady-state level	$\bar{\xi}$	0.20	Normalized
Persistence	ρ_ξ	0.96	Persistent fiscal episodes
CIR volatility scale	ν	0.40	Sets η -shock size $\nu\sqrt{\bar{\xi}}$
<i>Fiscal rule</i>			
Steady-state surplus	s^*	0.006	$s^*/b^* = \bar{r}$ (consistent SS real return)
Steady-state debt	b^*	0.60	60% debt/GDP
Consolidation coefficient	δ_b	0.30	Active fiscal ($\delta_b < 1$)
<i>New Keynesian block</i>			
NKPC slope	κ	0.10	Standard
IS real-rate sensitivity	ψ_r	0.50	
Natural output level	y_n	0.20	Pins SS level; does not affect IRFs
Discount factor	β	0.99	Quarterly
<i>Monetary policy</i>			
Taylor coefficient	ρ_π	0.85	Passive monetary ($\rho_\pi < 1$)
Steady-state nominal rate	i^*	0.03	$= \bar{r} + \pi^*$ (Fisher s.s.)
Inflation target	π^*	0.02	2% annual
<i>Bond duration</i>			
Roll-over fraction	ω	0.975	$D_{mac} = (1 + i^*)/(1 + i^* - \omega) \approx 18.7$ Q

Table 4: Derived Equilibrium Coefficients

Object	Formula	Eq. I	Eq. II	Interpretation
k_1	$b^*/(b^* + s^*)$	+0.990	+0.990	
k_2	$s^*/(b^* + s^*)$	+0.010	+0.010	$k_1 + k_2 = 1$
PT	$1 - \omega\rho_\pi$	+0.171	+0.171	Duration passthrough
γ	$\omega\rho_\pi/PT$	+4.839	+4.839	Effective duration
θ	from κ , App. A.2	+0.733	+0.733	Calvo; price duration $1/(1 - \theta) = 3.7$ Q
A_1	eq. (26)	-9.439	-9.439	Debt loading on r_t
A_2	eq. (33)	+0.000	+0.359	Debt loading on ξ_t
λ	eq. (32)	+0.037	+0.037	Price of real-rate risk
μ	eq. (32)	+0.000	-0.046	Price of fiscal risk
B_2	$-PT k_1 A_1$	+1.600	+1.600	π loading on r_{t+1}
B_4	$-PT k_1 A_2$	-0.000	-0.061	π loading on ξ_{t+1}
B_1	$-1 - B_2\rho_r$	-2.440	-2.440	π loading on r_t
$B_3 + B_4\rho_\xi$	eq. (38)	+0.000	+0.001	$\mathbb{E}_t[\pi_{t+1}]$ loading on ξ_t
$B_1 + B_2\rho_r$	Fisher check	-1.000	-1.000	Exact: -1
RP^ξ/ξ_t	$\mu k_1 A_2 \nu$	+0.000	-0.007	Fiscal risk premium per unit ξ_t
IRP at $\bar{\xi}$	$-(B_3 + B_4\rho_\xi)\bar{\xi}$	-0.000	-0.000	Inflation risk premium

static signs are parameter-free — but the quantitative statements in Section 5.4, and in particular whether the QE effect reverses sign, do. A calibration that identifies this block on fiscal-uncertainty

or survey-dispersion data is the most valuable next step.

Four sign results are central. First, $A_1 < 0$: a higher real rate lowers the real value of government debt, consistent with the FTPL channel whereby tighter financial conditions reduce the present value of future surpluses. Second, $A_2^* > 0$ in the Fiscal Sunspot Equilibrium: higher fiscal uncertainty raises real debt values. Third, $\mu < 0$ and $RP^\xi < 0$: bonds hedge fiscal uncertainty, so the fiscal risk premium is negative. Fourth, $B_1 < 0$ and $B_3 + B_4\rho_\xi > 0$: a higher real rate is deflationary (fiscal tightening via debt revaluation) while higher fiscal uncertainty raises expected inflation. The Fisher consistency condition $B_1 + B_2\rho_r = -1$ is satisfied to machine precision.

5 Results

We study the model's dynamics through deterministic impulse responses. Each IRF introduces a single one-standard-deviation shock at $t = 0$ and iterates the equilibrium laws of motion forward without further shocks, so that $\mathbb{E}_t[\pi_{t+1}]$ equals realized π_{t+1} exactly. Six variables are tracked: the two state variables (r_t, ξ_t) , the real value of government debt ($\log b_t$), inflation (π_t), output (y_t), and the inflation risk premium (40).

5.1 Real-Rate Shock (ε_{t+1})

Figure 1 reports the response to a one-standard-deviation ε shock ($\sigma = 0.005$, = 50 bp). Since the two shocks are orthogonal and A_1 does not depend on A_2 , the two equilibria produce *identical* responses: ξ_t and IRP_t remain identically zero throughout.

The real rate decays geometrically at rate $\rho_r = 0.90$. Real debt $\log b_t$ drops on impact by $A_1\sigma \approx -0.047$: a higher real rate depresses the present value of future surpluses. Through the FTPL valuation equation this revaluation raises inflation on impact by $B_2\sigma \approx +80$ bp (the stepping-on-a-rake effect). Inflation then turns negative as the contractionary backward channel ($B_1 < 0$) overtakes the impact revaluation, troughing near quarter 9 at ≈ -137 bp before recovering as r_t mean-reverts. Output rises on impact through the NKPC and then turns negative. The IRP is identically zero throughout, isolating the monetary transmission channel from fiscal uncertainty.

Duration attenuates the fundamental channel. The inflation impact is $B_2\sigma = -k_1A_1\sigma PT$, and $PT = 1 - \omega\rho_\pi$ strictly decreases in ω : longer-duration bonds absorb the revaluation as a capital gain rather than forcing an immediate price-level adjustment. At $\omega = 0$ the full one-period passthrough $-k_1A_1\sigma \approx 467$ bp operates; at the baseline $\omega = 0.975$ it is reduced to $PT \approx 17.1\%$ of that, ≈ 80 bp.

5.2 Fiscal Sunspot Shock (η_{t+1})

Figure 2 shows the response to a one-standard-deviation η shock (impact $v\sqrt{\bar{\xi}} \approx 0.179$). This is the paper's central result. In the Fundamental Equilibrium every panel except ξ_t is identically

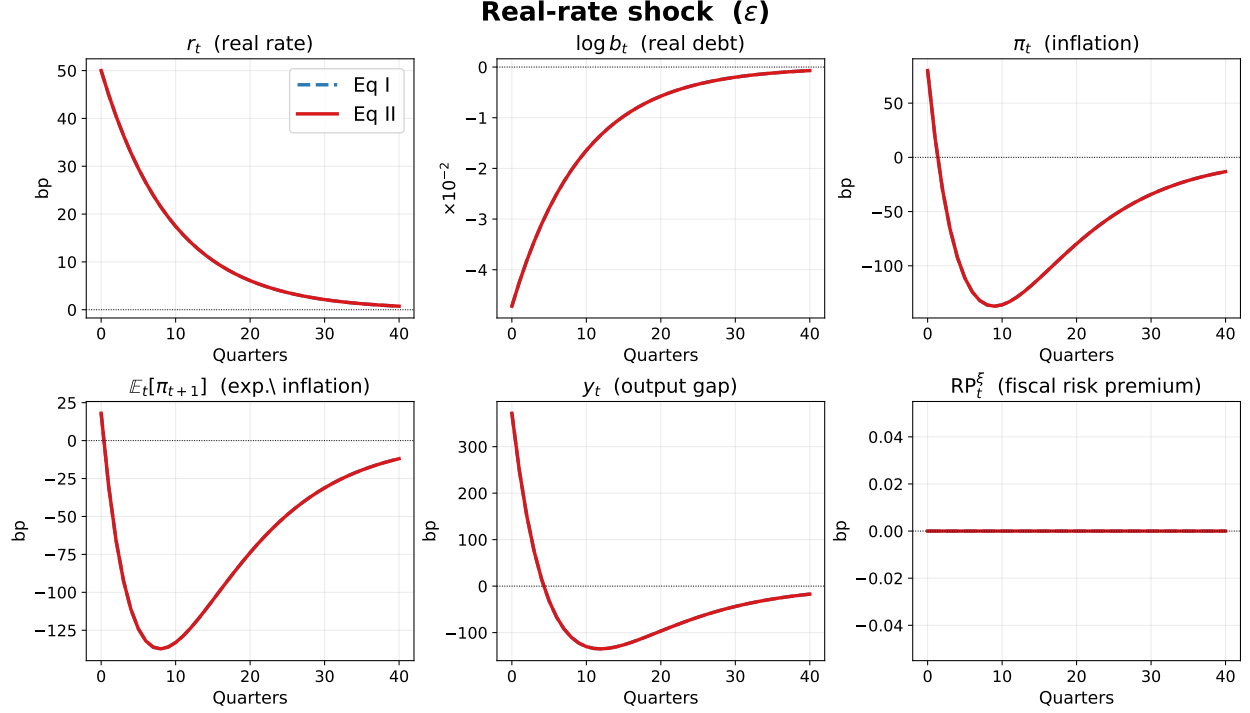


Figure 1: Impulse responses to a 1 s.d. real-rate shock (ε_{t+1} , $\sigma = 0.005$, impact = 50 bp). Blue dashed: Fundamental Equilibrium ($A_2 = 0$); red solid: Fiscal Sunspot Equilibrium ($A_2 > 0$). Passive monetary ($\rho_\pi = 0.85$), active fiscal ($\delta_b = 0.30$), $\omega = 0.975$ ($D_{mac} \approx 18.7$ Q). Deviations from steady state. The two equilibria coincide: ξ_t and the IRP are unaffected by the real-rate shock, by orthogonality.

zero: the shock raises fiscal uncertainty but has no pricing implications. In the Fiscal Sunspot Equilibrium the same shock activates the debt revaluation channel.

Debt revaluation. Fiscal uncertainty rises on impact and decays at rate $\rho_\xi = 0.96$. Because $A_2^* > 0$, real debt is immediately revalued upward: $\log b_t$ rises by $A_2^* v \sqrt{\bar{\xi}} \approx +0.069$. Investors who price the η shock ($\mu < 0$) treat the bond as a hedge and bid up its value.

FTPL inflation channel. The FTPL translates the higher real debt value into price-level dynamics through two competing forces in (36). On impact, the $B_4 < 0$ loading produces *deflation* of ≈ -117 bp: a larger real liability requires the current price level to fall to restore FTPL consistency. Thereafter the $B_3 + B_4 \rho_\xi > 0$ loading feeds rising expected inflation through the passive Taylor rule, eventually dominating. The response is hump-shaped, peaking near quarter 14 at $\approx +137$ bp before mean-reverting.

Output and the risk premium. Output falls sharply on impact, with the initial deflation dominating the NKPC. The IRP satisfies $IRP_t = -(B_3 + B_4 \rho_\xi) \xi_t < 0$, falling by ≈ 35 bp on impact: nominal bonds reward investors precisely when fiscal uncertainty is high. The real rate is unchanged throughout, confirming that the η shock carries no macro-fundamental content in either

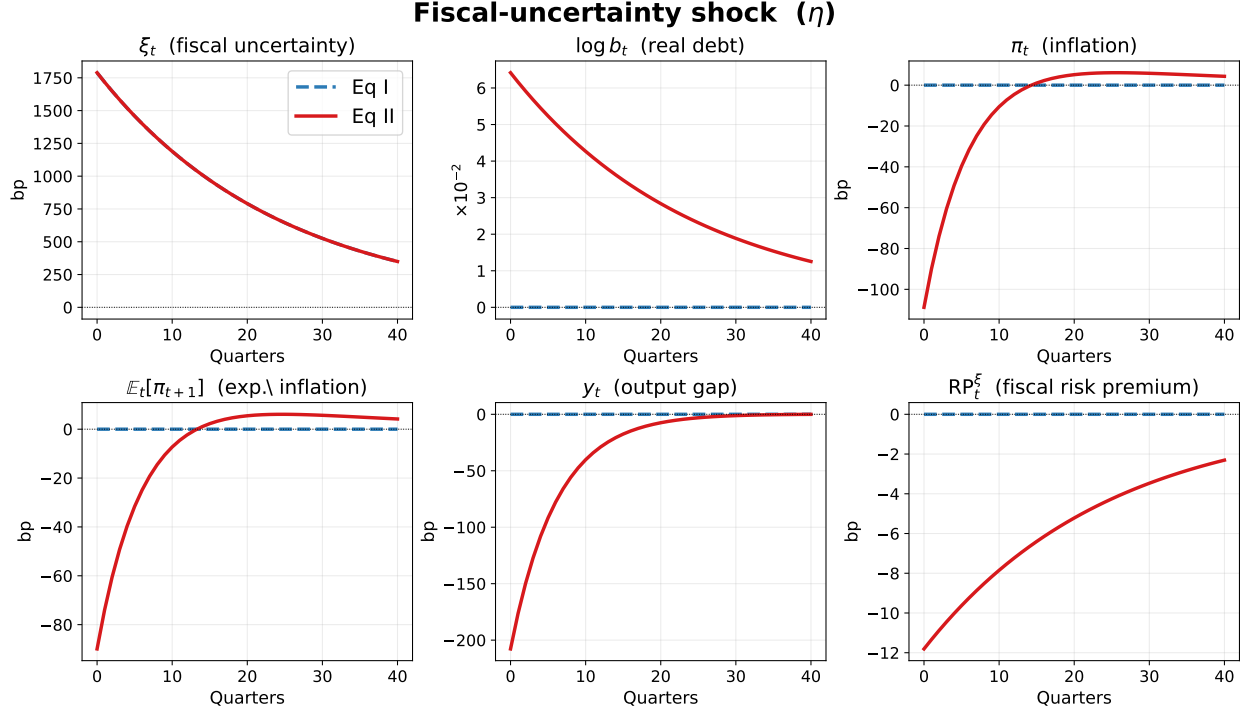


Figure 2: Impulse responses to a 1 s.d. fiscal sunspot shock (η_{t+1} , impact $v\sqrt{\xi} \approx 0.179$). Blue dashed: Fundamental Equilibrium ($A_2 = 0$, all panels except ξ_t identically zero); red solid: Fiscal Sunspot Equilibrium ($A_2 > 0$). Passive monetary ($\rho_\pi = 0.85$), active fiscal ($\delta_b = 0.30$), $\omega = 0.975$. Deviations from steady state.

equilibrium. The model cannot predict which equilibrium is realized; it characterizes the two self-consistent outcomes and their distinct empirical signatures.

5.3 Debt Maturity and the Asymmetric Duration Effect

Because ω enters the equilibrium only through PT and through the way PT feeds A_2^* , the effect of debt maturity on each channel can be signed in closed form. Appendix A.9 derives the derivatives below and verifies each against a central finite difference of the solver, with a maximum relative error of 8.1×10^{-11} . Table 5 reports the values.

The two channels move in *opposite* directions:

- **Fundamental channel: dampened by duration.** $\partial B_2 / \partial \omega = k_1 A_1 \rho_\pi < 0$. Longer maturity shrinks the passthrough of real-rate shocks to inflation, because PT falls.
- **Sunspot channel: amplified by duration.** $\partial A_2^* / \partial \omega > 0$ and $\partial (B_3 + B_4 \rho_\xi) / \partial \omega > 0$. Longer maturity raises the fiscal-uncertainty loading in bond values and the expected-inflation sensitivity to ξ_t , so the safe-haven IRP deepens: $\partial \text{IRP}_t / \partial \omega < 0$.

This asymmetry is the paper's central maturity result. It arises because PT enters the two channels

Table 5: Sensitivity of equilibrium objects to bond duration ω (Fiscal Sunspot Equilibrium, baseline $\omega = 0.975$).

Object	Symbol	Value	$\partial/\partial\omega$
Passthrough factor	$PT = 1 - \omega\rho_\pi$	+0.1713	-0.8500
Effective duration	γ	+4.8394	+28.9840
Fiscal debt loading	A_2^*	+0.3587	+0.7767
Real-rate risk price	λ	+0.0372	-0.0630
Fiscal risk price	μ	-0.0465	+0.1538
π loading on r_{t+1}	B_2	+1.6005	-7.9439
π loading on ξ_{t+1}	B_4	-0.0608	+0.1702
π loading on r_t	B_1	-2.4404	+7.1495
$\mathbb{E}_t \pi_{t+1}$ on ξ_t	$B_3 + B_4\rho_\xi$	+0.0014	-0.0086
IRP per unit ξ_t	$-(B_3 + B_4\rho_\xi)$	-0.0014	+0.0086

differently: it multiplies the fundamental loading B_2 directly, but in the sunspot channel it enters the fixed point (33) nonlinearly, so a lower PT raises A_2^* and partly offsets its own dampening effect.

Implication for quantitative easing. Quantitative easing shortens the maturity of the consolidated government balance sheet, a fall in ω . The comparative statics imply that QE operates asymmetrically on the two equilibria: it *raises* the sensitivity of inflation to fundamental real-rate shocks (through a higher PT , hence larger $|B_2|$) while *compressing* the self-fulfilling fiscal channel (through a lower A_2^*). Figure 4 quantifies the second effect: halving the roll-over fraction from $\omega_H = 0.975$ to $\omega_L = 0.4875$ lowers A_2^* and $B_3 + B_4\rho_\xi$, cutting the peak inflation response to a fiscal sunspot shock from +137 bp to +62 bp, a reduction of about 55%.

5.4 How Much QE Disinflates Depends on the Equilibrium

The comparative statics above hold the equilibrium fixed. The sharper question is what happens to the *level* of expected inflation when maturity changes, because there the two equilibria disagree quantitatively — and, for a large enough fiscal loading, in sign.

Consider an MIT shock that shortens maturity from $\omega_H = 0.975$ to $\omega_L = \omega_H - 0.02$, reverting at rate $\rho_\omega = 0.90$. Because ω enters only through PT and through PT 's effect on A_2^* , the impact response of $\mathbb{E}_t[\pi_{t+1}]$ decomposes into four channels (Appendix A.10): the two conventional term-premium channels of Corhay et al. [2023] — a *quantity* effect $TP \Delta\gamma$ and a *repricing* effect $\gamma \Delta TP$ — plus two balance-sheet channels operating through the FTPL constant condition (69): an A_0 *hedge* channel running through $\lambda(\omega)$, and an A_2 *coordination* channel that exists only when fiscal uncertainty is priced. Table 6 reports the decomposition.

Three things stand out. First, the conventional term-premium channels are *small* — together under 1.2 bp in either equilibrium — because the quantity and repricing effects nearly offset, as the product-rule structure of γTP implies. The accounting of Corhay et al. [2023] captures only these. Second, the A_0 hedge channel is -12.9 bp and is *identical across equilibria*: it runs entirely through

Sensitivity of equilibrium objects to bond duration ω (Fiscal Sunspot Equilibrium; baseline $\omega = 0.975$ dashed)

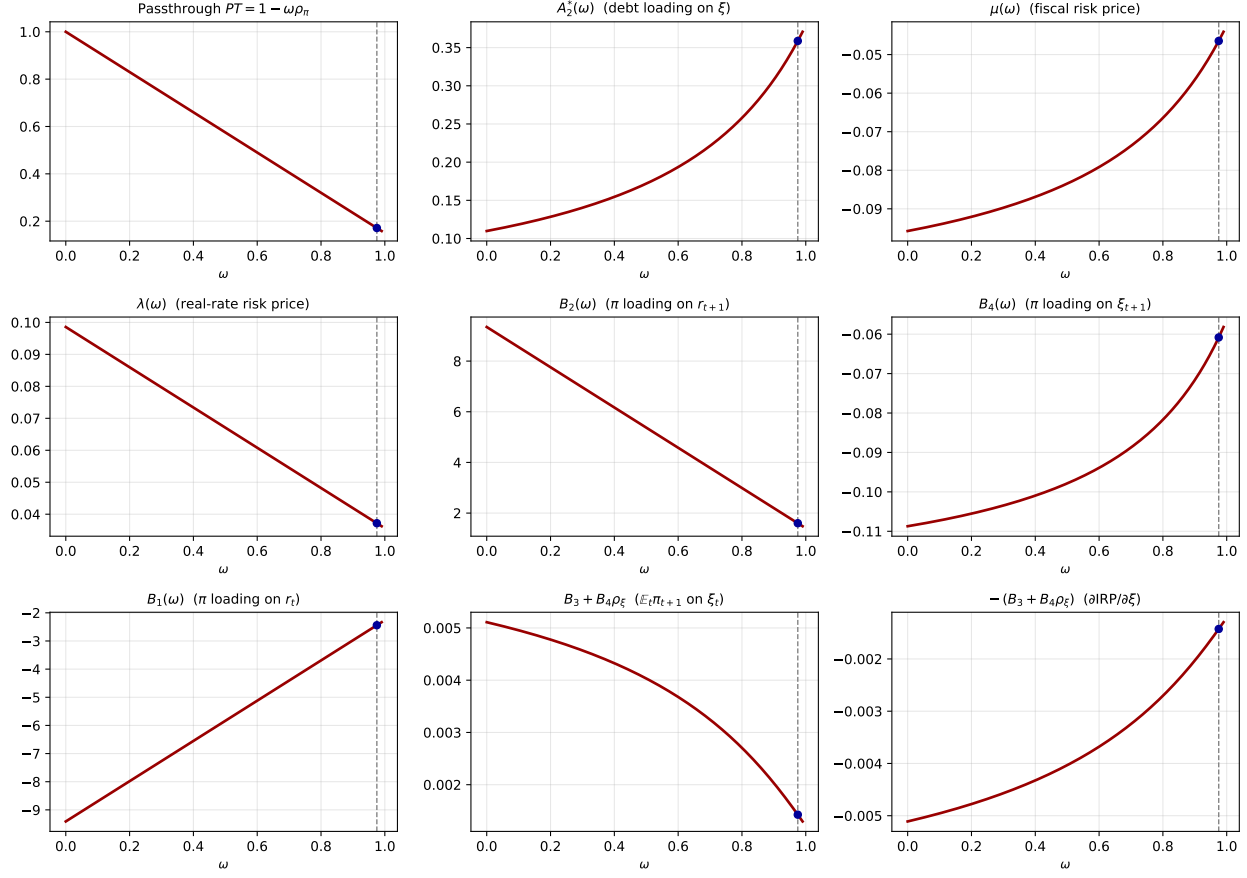


Figure 3: Comparative statics of the equilibrium in the bond-duration parameter ω (Fiscal Sunspot Equilibrium). Each panel plots one object over $\omega \in [0, 0.99]$; the dashed line and dot mark the baseline $\omega = 0.975$. The passthrough factor PT , fundamental loadings (B_2, B_1), and the real-rate risk price λ shrink in magnitude with duration, while the fiscal debt loading A_2^* and the expected-inflation sensitivity $B_3 + B_4\rho_\xi$ rise, so the safe-haven inflation risk premium deepens.

Table 6: Impact effect of a maturity shortening ($\Delta\omega = -0.02$, $\rho_\omega = 0.90$) on expected inflation, in basis points. The A_2 coordination channel is identically zero in the Fundamental Equilibrium and is what attenuates the total in the Fiscal Sunspot Equilibrium.

Channel	Source	Eq. I	Eq. II
Term-premium quantity	TP $\Delta\gamma$	+0.60	+0.12
Term-premium repricing	$\gamma \Delta TP$	-1.15	-0.63
A_0 balance-sheet hedge	$\lambda \rightarrow A_0$	-12.90	-12.90
A_2 coordination channel	$A_2 \rightarrow A_0, \Delta A_2 \bar{\xi}$	+0.00	+32.13
Total	$\partial \mathbb{E}_t[\pi_{t+1}]$	-13.45	+18.72

$\lambda(\omega)$ and does not involve A_2 . On its own it makes maturity shortening disinflationary. Third, the A_2 coordination channel contributes +5.9bp, but *only* in the Fiscal Sunspot Equilibrium. At the

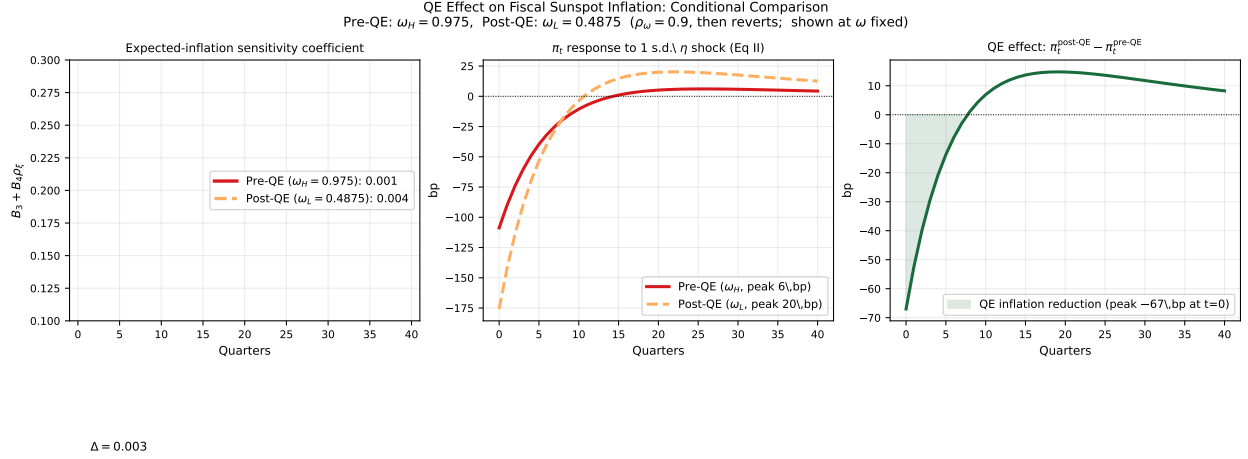


Figure 4: Maturity shortening compresses the fiscal-sunspot inflation response. *Left*: the expected-inflation sensitivity $B_3 + B_4\rho_\xi$ at $\omega_H = 0.975$ versus $\omega_L = 0.4875$. *Centre*: inflation response to a 1 s.d. η shock in the long-maturity (red) and short-maturity (orange dashed) Fiscal Sunspot Equilibrium. *Right*: difference path. Fiscal uncertainty is normalized, $\xi = 0.20$.

baseline it does not overturn the hedge channel, but it cuts the disinflationary effect of the same policy roughly in half, from -13.4 bp to -7.9 bp.

The policy implication is that the effectiveness of a maturity operation is not a structural constant. With identical fundamentals, rules, and states, the same shortening delivers markedly different disinflation depending on whether investors have coordinated on pricing fiscal uncertainty — and, as the next paragraph shows, the effect can change sign outright once the fiscal loading is large enough.

When does the effect reverse sign? The coordination channel scales with A_2^* , so whether it merely attenuates the hedge channel or overturns it is a quantitative question. Table 7 reports the sensitivity. At the baseline, $A_2^* = 0.21$ and the effect is attenuated but not reversed. A reversal requires roughly $A_2^* \gtrsim 0.3$, which obtains for a steeper Phillips curve ($\kappa \geq 0.2$) or a less volatile fiscal-uncertainty process ($\nu \leq 0.2$). Across a grid of 320 parameter combinations spanning ω , κ , ν and ρ_π , a sign reversal occurs in 62.

This is a weaker claim than the attenuation result, and it should be read as such: the sign reversal is a possibility the model admits over a sizable region of the parameter space, not a prediction of the baseline calibration. It is also sensitive to precisely the parameter, ν , that the fiscal-uncertainty block is least well identified on — see the discussion of the calibration in Section 4.

5.5 Does the Second Equilibrium Survive?

The Fiscal Sunspot Equilibrium relies on a chain running from the Phillips curve to the SDF to the bond-pricing condition. Table 8 asks which links are essential, restricted to modifications the

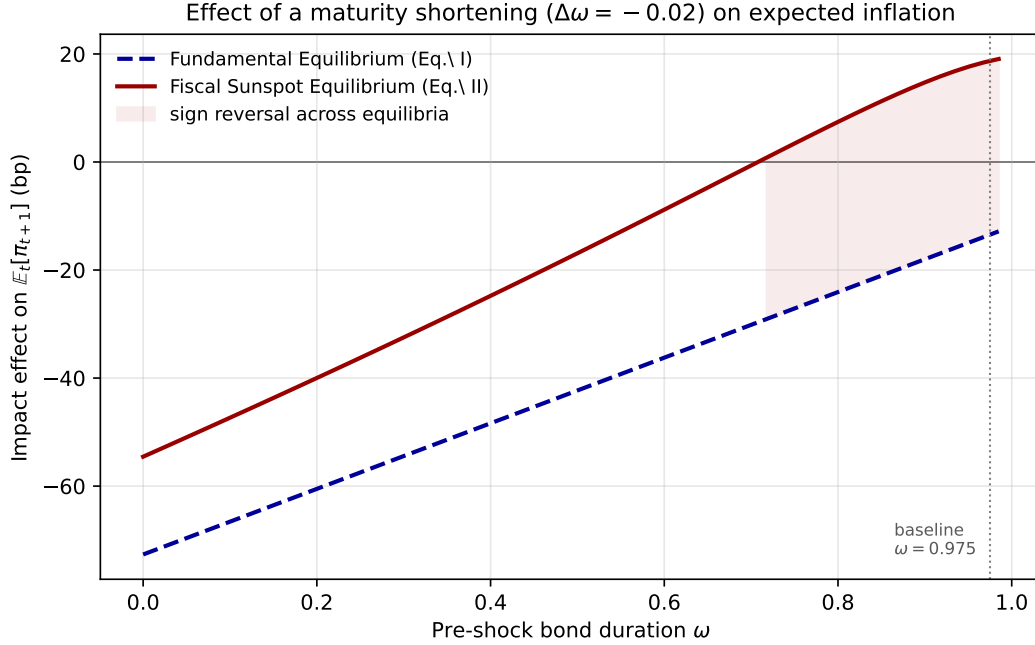


Figure 5: Impact effect of a maturity shortening on expected inflation, as a function of the pre-shock duration ω , in both equilibria. Blue dashed: Fundamental Equilibrium; red solid: Fiscal Sunspot Equilibrium. The gap between the curves is the A_2 coordination channel, which widens with duration because A_2^* does.

present model can actually evaluate.

Two results deserve emphasis. *Long-maturity debt is not necessary.* At $\omega = 0$ the second root survives with $A_2^* = 0.142 > 0$: the multiplicity does not rest on duration, which only amplifies it. This distinguishes the mechanism from the debt-dilution multiplicity of [Aguiar and Amador \[2020\]](#), where long bonds are indispensable. *Sticky prices matter for the mechanism, not for existence.* As $\kappa \rightarrow \infty$ the risk price $\mu \rightarrow 0$, so the self-fulfilling SDF loop switches off — yet a second root persists, now sustained purely by the Jensen term in the CIR variance ($\frac{1}{2}k_1^2 A_2 v^2$ in (24)). The flexible-price multiplicity is therefore a different and weaker object: it no longer involves coordinated pricing of fiscal risk. Active monetary policy destroys the equilibrium outright, since $\rho_\pi > 1$ renders the inflation recursion explosive.

A policymaker who believes the economy is in the Fiscal Sunspot Equilibrium can therefore use maturity shortening to compress self-fulfilling premia, at the cost of greater sensitivity to fundamental shocks. The trade-off has no counterpart in the Fundamental Equilibrium, where the fiscal channel is absent and only the first effect operates.

Table 7: Sensitivity of the QE effect. Entries are the impact effect on $\mathbb{E}_t[\pi_{t+1}]$ (bp) in each equilibrium; † marks the baseline value. A sign reversal requires Eq. I $< 0 <$ Eq. II.

Parameter	Value	Eq. I	Eq. II	A_2^*	Sign reversal?
ω	0.9	−18.0	+14.7	0.308	✓
	0.95	−15.0	+17.6	0.340	✓
	0.975 [†]	−13.4	+18.7	0.359	✓
	0.99	−12.5	+19.2	0.371	✓
ρ_π	0.5	−41.2	−36.0	0.090	−
	0.7	−31.1	−13.6	0.185	−
	0.85 [†]	−13.4	+18.7	0.359	✓
	0.95	−3.1	+16.4	0.534	✓
κ	0.05	−26.9	+4.6	0.257	✓
	0.1 [†]	−13.4	+18.7	0.359	✓
	0.2	−6.7	+20.0	0.445	✓
	0.5	−2.7	+14.1	0.520	✓
δ_b	0.1	−10.1	+17.3	0.373	✓
	0.3 [†]	−13.4	+18.7	0.359	✓
	0.5	−19.6	+21.4	0.344	✓
	0.9	−105.7	+61.8	0.316	✓
ρ_ξ	0.9	<i>Eq. II does not exist</i>			
	0.94	−13.4	+54.4	0.497	✓
	0.96 [†]	−13.4	+18.7	0.359	✓
	0.98	−13.4	−3.3	0.212	−
ν	0.2	−13.4	+115.2	1.435	✓
	0.3	−13.4	+43.7	0.638	✓
	0.4 [†]	−13.4	+18.7	0.359	✓
	0.6	−13.4	+0.8	0.159	✓

Table 8: Does the Fiscal Sunspot Equilibrium survive structural modifications?

Modification	Eq. II survives?	At baseline	Why
Baseline	✓ yes	$A_2^* = 0.359, \mu = -0.046$	Self-fulfilling loop operative
Short-term debt only ($\omega = 0$)	✓ yes	$A_2^* = 0.110, \mu = -0.096$	Survives but weaker: no duration amplification
Flexible prices ($\kappa \rightarrow \infty$)	✓ weaker	$A_2^* = 0.593, \mu = -1.0e - 08$	Root survives via CIR Jensen term, not the SBC
Active Taylor rule ($\rho_\pi > 1$)	× no	—	$\rho_\pi > 1$: π recursion explosive, $PT < 0$

6 Optimal Debt Maturity

The comparative statics of Section 5.3 treat ω as a parameter the government moves once. This section asks instead what maturity policy the government *should* follow, and distinguishes two environments that give different answers: an unanticipated (MIT) change in maturity, and a maturity *rule* that agents anticipate and price.

6.1 The Government's Objective

Two ingredients are needed: a welfare criterion and a reason not to make debt arbitrarily long.

Welfare. Taking a second-order approximation to the household's objective (3) around the efficient steady state, and using the Calvo price dispersion implied by (11), gives the standard quadratic criterion (Appendix A.7):

$$\mathcal{W} \simeq -\frac{1}{2} \mathbb{E}_0 \sum_{t \geq 0} \beta^t \left[\frac{\epsilon_p}{\kappa} \pi_t^2 + (1 + \psi) \tilde{y}_t^2 \right] + \text{t.i.p.}, \quad (41)$$

where $\tilde{y}_t$ is the output gap. Two features matter. First, the relative weight on output is *not* a free parameter: it is

$$\alpha \equiv \frac{(1 + \psi) \kappa}{\epsilon_p}, \quad (42)$$

pinned by the Frisch elasticity, the Phillips slope and the demand elasticity, and equal to 0.033 at the calibration. Second, ϵ_p — which entered the model only through the risk wedge Λ_t — now also sets the weight on inflation, so the same parameter disciplines both.

Liquidity services. With (41) alone the government would lengthen maturity without limit, since longer debt damps the passthrough of every shock. Short debt must therefore be worth something on its own. Following the convenience-yield tradition, let the household derive services from the fraction of debt maturing next period, $\ell_t \equiv (1 - \omega_t) b_t$, adding $\chi v(\ell_t)$ to period utility with v increasing and concave, $v(\ell) = \ell^{1-\zeta}/(1 - \zeta)$. The government's problem is then

$$\min_{\{\omega_t\}} \frac{1}{2} \left[\frac{\epsilon_p}{\kappa} \text{Var}(\pi) + (1 + \psi) \text{Var}(\tilde{y}) \right] - \chi \mathbb{E}[v(\ell_t)], \quad \omega_t \in [0, 1), \quad (43)$$

subject to the equilibrium conditions of Section 3. This is a Ramsey problem in the maturity instrument alone: monetary and fiscal rules are held fixed, so the government is not choosing the price level, only how much of each shock the price level absorbs.

6.2 Maturity Rebalancing Is Value-Neutral

Before solving, one lemma makes the whole exercise tractable.

Lemma 1. *Suppose the government rebalances its portfolio each period at market prices, choosing ω_t at date t . Then rebalancing transfers no value, and the bond-return innovation is*

$$r_{g,t+1} - \mathbb{E}_t[r_{g,t+1}] = -\gamma_t (\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]) = \omega_t \rho_\pi (r_{s,t+1} - \mathbb{E}_t[r_{s,t+1}]), \quad (44)$$

with $\gamma_t = \omega_t \rho_\pi / (1 - \omega_t \rho_\pi)$ and $PT_t = 1/(1 + \gamma_t) = 1 - \omega_t \rho_\pi$, both $\mathcal{F}_t$ -measurable.

The second equality uses $\gamma_t PT_t = \omega_t \rho_\pi$ identically. The content of the lemma is that a state-contingent maturity policy does *not* add a revaluation term to the bond return: swapping long for short debt at market prices leaves b_t unchanged. Maturity policy operates only by changing the portfolio's *future* risk exposure through PT_t , never by changing its current value. This is what separates the anticipated rule from the MIT shock, where the change in ω is precisely *not* priced and therefore does revalue the balance sheet.

A caveat on the affine class. If ω_t responds to the state then PT_t does too, and the conditional variance of inflation, $PT_t^2 k_1^2 (A_1^2 \sigma^2 + A_2^2 v^2 \xi_t)$, is no longer affine in ξ_t . No non-trivial state-contingent rule preserves exact affineness. The analysis below is therefore a perturbation: it is exact to first order in the policy slope, which is the standard treatment of policy rules in affine models, and it is the reason the results take the form of a marginal condition rather than a global solution.

6.3 An Anticipated Maturity Rule

Let the government commit to a rule that agents observe and price,

$$\omega_t = \bar{\omega} + \omega_\xi (\xi_t - \bar{\xi}), \quad (45)$$

with ω_ξ the policy slope: $\omega_\xi > 0$ lengthens maturity when fiscal uncertainty is high, $\omega_\xi < 0$ shortens it. By Lemma 1 the inflation innovation is $\Delta\pi_{t+1} = -PT_t k_1 (A_1 \sigma \varepsilon_{t+1} + A_2 v \sqrt{\xi_t} \eta_{t+1})$, so its unconditional variance depends on the rule through two moments,

$$\mathbb{E}[PT_t^2] = \bar{PT}^2 + \rho_\pi^2 \omega_\xi^2 \text{Var}(\xi), \quad \mathbb{E}[PT_t^2 \xi_t] = \bar{PT}^2 \bar{\xi} - 2\bar{PT} \rho_\pi \omega_\xi \text{Var}(\xi) + O(\omega_\xi^2). \quad (46)$$

These two moments carry the entire trade-off, and they pull in opposite directions:

$$\frac{\mathbb{E}[\text{var}_t(\pi_{t+1})]}{k_1^2} = \underbrace{A_1^2 \sigma^2 \left[\bar{PT}^2 + \rho_\pi^2 \omega_\xi^2 \text{Var}(\xi) \right]}_{\text{real-rate channel: convex, penalises any state-contingency}} + \underbrace{A_2^2 v^2 \left[\bar{PT}^2 \bar{\xi} - 2\bar{PT} \rho_\pi \omega_\xi \text{Var}(\xi) \right]}_{\text{sunspot channel: linear, rewards lengthening}}. \quad (47)$$

The economics is transparent. The η shock arrives with conditional volatility $v\sqrt{\xi_t}$: it is *large exactly when ξ_t is large*. Lengthening maturity in those states lowers PT_t and so absorbs the shock as a capital gain rather than a price-level move. That is a genuine hedge and it is why the sunspot term is linear and decreasing in ω_ξ . The ε shock, by contrast, has constant volatility; making PT_t move with ξ_t simply adds variance to its passthrough without any offsetting benefit, which is why the real-rate term is convex in ω_ξ and minimized at $\omega_\xi = 0$.

The optimal slope. Differentiating (47) and setting the derivative to zero gives, for the inflation-

stabilization motive alone,

$$\omega_\xi^* = \frac{A_2^2 v^2 \overline{PT}}{A_1^2 \sigma^2 \rho_\pi} \quad (48)$$

This is the equation linking maturity to the underlying state. It says the optimal *responsiveness* of duration to fiscal uncertainty is the ratio of the two shocks' contributions to inflation risk, scaled by the passthrough factor. It is positive whenever $A_2 > 0$: *in the Fiscal Sunspot Equilibrium the government should lengthen maturity when fiscal uncertainty rises, and in the Fundamental Equilibrium it should not respond at all*, since $A_2 = 0$ sets $\omega_\xi^* = 0$. State-contingent maturity policy is valuable only when fiscal risk is priced.

The constraint binds. At the calibration $\omega_\xi^* = 1.86$, so a one-standard-deviation rise in ξ_t would call for $\Delta\omega = 1.19$ — far outside $[0, 1)$. The sunspot channel outweighs the real-rate channel by a factor of 1.9 in (48), and with $\bar{\omega} = 0.975$ already near the consol bound the unconstrained optimum is infeasible. Optimal maturity policy is therefore a *corner*: lengthen to the longest available maturity when fiscal uncertainty is high. What pulls policy back from the corner is the liquidity motive, to which we now turn.

6.4 The Liquidity Trade-off

Differentiating the liquidity term in (43) under the rule (45), and using $\partial\ell_t/\partial\omega_\xi = -(\xi_t - \bar{\xi})b_t$,

$$\frac{\partial}{\partial\omega_\xi} \mathbb{E}[v(\ell_t)] = -\text{Cov}(v'(\ell_t) b_t, \xi_t) = -(1 - \bar{\omega})^{-\zeta} \text{Cov}(b_t^{1-\zeta}, \xi_t). \quad (49)$$

Because $A_2 > 0$, real debt b_t is increasing in ξ_t , so the covariance is positive whenever $\zeta < 1$ and the derivative is negative. The liquidity motive therefore pushes ω_ξ *down*, opposing the stabilization motive, and it does so for an intuitive reason: the state in which the government would like to lengthen is exactly the state in which its debt — and hence the stock of liquidity services it supplies — is largest, so lengthening forgoes the most liquidity precisely when the most is at stake. At $\zeta = 1$ the two effects cancel exactly and the liquidity motive is silent; for $\zeta > 1$ it reinforces lengthening.

Combining, the interior first-order condition for the slope is

$$\underbrace{\frac{\epsilon_p}{\kappa} k_1^2 \rho_\pi \text{Var}(\xi) \left[A_1^2 \sigma^2 \rho_\pi \omega_\xi - A_2^2 v^2 \overline{PT} \right]}_{\text{inflation stabilization}} + \underbrace{(1 + \psi) \frac{\partial \text{Var}(\tilde{y})}{\partial \omega_\xi}}_{\text{output}} + \underbrace{\chi (1 - \bar{\omega})^{-\zeta} \text{Cov}(b_t^{1-\zeta}, \xi_t)}_{\text{liquidity}} = 0, \quad (50)$$

which reduces to (48) when $\chi = 0$ and the output term is neglected. Equation (50) is the sense in which the model delivers a maturity *rule* rather than a maturity *number*: ω_ξ^* is a function of the deep parameters governing how each shock reaches the price level and how valuable short debt is at the margin.

6.5 Contrast: the MIT Shock

The MIT experiment of Section 5.4 is the opposite informational case: agents do not anticipate the change in ω , so it is not in their pricing functions when the change occurs. Three differences follow, and they are worth separating because they are easy to conflate.

1. **Revaluation.** Under the anticipated rule, Lemma 1 says rebalancing moves no value. Under the MIT shock the change in ω reprices A_0 and A_2 , and that repricing is the entire content of Table 6: the -12.9 bp hedge channel and the $+5.9$ bp coordination channel are both revaluation effects that the anticipated rule does not generate.
2. **Which margin matters.** The MIT shock operates on the *level* of ω and its effect is a one-time inflation innovation. The anticipated rule operates on the *slope* ω_ξ and its effect is on second moments. A policy can be disinflationary in the first sense and stabilizing in the second, or the reverse.
3. **Time consistency.** Because the anticipated rule is priced, the government faces the usual temptation to deviate once beliefs are formed: having induced agents to price a high- ω_ξ rule, it would prefer ex post to shorten and harvest liquidity. The MIT experiment is silent on this by construction. A full treatment would need to distinguish commitment from discretion, which we leave for future work.

The practical reading is that the two experiments answer different questions. The MIT shock asks what happens when maturity changes unexpectedly, and its answer depends on which equilibrium investors have coordinated on. The anticipated rule asks how maturity *should* covary with fiscal conditions, and its answer — lengthen when fiscal uncertainty rises, subject to the liquidity cost — holds in the Fiscal Sunspot Equilibrium and is vacuous in the Fundamental Equilibrium.

7 What Coordinates Beliefs?

Any multiplicity result invites the question of why the economy would be in one equilibrium rather than the other. The model is deliberately silent on selection — both equilibria are fully rational and neither is Pareto-dominant — but the sunspot is not mystical, and it is worth being concrete about what could play its role.

The object ξ_t is a *coordination state*: a publicly observed variable that investors may or may not treat as relevant for pricing. Candidates that would make the Fiscal Sunspot Equilibrium more likely include episodes that make the fiscal position salient without changing it — debt-ceiling standoffs and budget confrontations, sovereign rating actions, prominent official warnings about

debt sustainability, or a shift in public narrative toward fiscal dominance. Changes in the investor base matter too: a shift from price-insensitive official holders toward leveraged or mark-to-market investors raises the weight on risk-based pricing. So does safe-asset scarcity, which makes the hedging value of government debt more valuable at the margin. Finally, QE announcements are themselves candidates, because they make the maturity structure of the consolidated balance sheet salient — which, given Section 5.4, is exactly the margin that determines how much the coordination channel is worth.

Two observations sharpen this. First, because the two equilibria differ in risk premia rather than in average inflation, selection is in principle detectable in asset prices without requiring a long sample of realized inflation. Second, the comparative statics of Section 5.3 imply that the coordination channel is worth more when debt maturity is long, so lengthening maturity should make coordination on the Fiscal Sunspot Equilibrium more attractive — a form of policy feedback into equilibrium selection that a full treatment would need to endogenize. Modelling adaptive learning over the risk price μ , and asking which fixed point is locally stable, is the natural next step and is left for future work.

8 Empirical Implications

The model is not estimated here, but it delivers sharp qualitative predictions that distinguish the two equilibria and are testable with standard term-structure data.

1. **Fiscal uncertainty shocks move inflation risk premia only in the Fiscal Sunspot Equilibrium.** In the Fundamental Equilibrium, innovations to a fiscal uncertainty proxy should be orthogonal to inflation risk premia and to real debt valuations; in the Fiscal Sunspot Equilibrium they move both. Regime-dependence in this relationship, rather than its average strength, is the object of interest.
2. **The sign of the bond-market response identifies the equilibrium.** If fiscal news raises nominal bond prices and *lowers* inflation compensation, that is the deflation-hedge signature of the Fiscal Sunspot Equilibrium. If it raises inflation compensation and yields together, the relevant channel is fiscal stress, which is outside this model.
3. **QE should have state-dependent inflation effects.** By Section 5.4, maturity operations should look inflationary precisely when fiscal risk is already strongly priced, and disinflationary otherwise. The prediction is a sign flip conditional on the pricing state, not a magnitude difference.
4. **Debt maturity interacts with fiscal-risk pricing.** Since $\partial A_2^*/\partial \omega > 0$, the loading of bond valuations on fiscal uncertainty should be stronger in periods and countries with longer

average maturity of privately held debt.

Natural empirical counterparts include TIPS breakevens and inflation swaps, decomposed inflation risk premia and Treasury term premia, fiscal uncertainty and news indices, the maturity structure of privately held Treasury debt, high-frequency QE announcement surprises, and survey dispersion about inflation and fiscal policy — the last being a natural proxy for the coordination state itself. Assembling these into a test of prediction 2, which requires only a sign, is the most direct route to discriminating between the two equilibria.

9 Conclusion

This paper shows that the Fiscal Theory of the Price Level generates a novel form of inflation indeterminacy: not a multiplicity of price-level *paths*, but a multiplicity of inflation *risk premia*. The fiscal valuation equation admits two rational-expectations equilibria consistent with the same structural parameters and current fundamentals.

In the *Fundamental Equilibrium*, the non-fundamental fiscal uncertainty shock η is entirely unpriced. Bond prices load only on the real rate, the stochastic discount factor carries no fiscal risk premium, and η has no equilibrium effect on inflation, output, or asset prices. The inflation risk premium is constant and driven solely by real-rate covariance.

In the *Fiscal Sunspot Equilibrium*, the same shock η is self-fulfilling through a *debt revaluation channel*. When investors price fiscal uncertainty ($\mu < 0$), the real value of government debt rises with ξ_t ($A_2 > 0$), so nominal bonds hedge fiscal risk and carry a negative risk premium. The Fiscal Theory then requires the current price level to adjust: the impact loading $B_4 < 0$ delivers deflation, after which the expected-inflation loading $B_3 + B_4\rho_\xi > 0$ raises inflation with ξ_t . But if inflation responds to η shocks, the stochastic discount factor must load on η to price inflation risk — validating $\mu < 0$ and $A_2 > 0$ and closing the loop. The non-fundamental shock becomes fundamental precisely because the pricing equation makes it so.

A key structural feature of the model is orthogonality: the η shock operates independently of the common macro shock ε that governs real rates and the output gap. In the Fundamental Equilibrium, this orthogonality is trivially satisfied — η is irrelevant. In the Fiscal Sunspot Equilibrium, η generates a time-varying inflation risk premium and inflation dynamics that are orthogonal to real interest rate movements. This separation is an exact analytical property of the model, not an approximation.

A second result concerns government debt maturity. The roll-over parameter ω enters the equilibrium through a single passthrough factor $PT = 1 - \omega\rho_\pi$, which acts on the two channels in opposite directions: it multiplies the fundamental inflation loading directly, but enters the denominator of the fixed point determining the fiscal loading A_2 . Longer maturity therefore

dampens the fundamental channel while amplifying the self-fulfilling one and deepening the safe-haven inflation risk premium. Maturity shortening compresses the sunspot channel at the cost of greater exposure to fundamental shocks — a trade-off that exists only when investors have coordinated on pricing fiscal uncertainty.

The model is deliberately parsimonious. The regime requirements are transparent: passive monetary policy ($\rho_\pi < 1$) and active fiscal policy ($\delta_b < 1$) are jointly necessary for the FTPL valuation channel to operate, and both are required for the Fiscal Sunspot Equilibrium to exist. The affine structure delivers closed-form expressions for all equilibrium objects, and the self-fulfilling mechanism is fully characterized by the quadratic (33) and the sign of S in (34).

Several extensions are natural. First, an equilibrium-selection or learning mechanism that determines which equilibrium obtains, and how transitions between them occur, would make the indeterminacy empirically operational. Second, the nominal term structure of Appendix A.11 implies distinct yield-curve signatures for the two equilibria, which suggests an identification strategy in term-structure data. Third, relaxing the assumption that the fiscal uncertainty shock is orthogonal to fundamentals would allow the model to speak to episodes in which fiscal stress and macroeconomic risk move together.

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A Appendix

This appendix collects the derivations behind the results stated in Section 2. Throughout, bond maturity enters only through the effective duration and passthrough factor of Section 3.1,

$$\gamma = \frac{\omega \rho_\pi}{1 - \omega \rho_\pi}, \quad PT = \frac{1}{1 + \gamma} = 1 - \omega \rho_\pi,$$

and the forward-adjusted NKPC slope $\tilde{\kappa} = \kappa / (1 - \beta \rho_\pi)$.

A.1 Fiscal Rule Log-Linearization

The surplus rule is $s_{t+1} = s^*(b_t/b^*)^{\delta_b}$, so $\log s_{t+1} = \log s^* + \delta_b(\log b_t - \log b^*)$. The flow budget constraint $b_{t+1} + s_{t+1} = (1 + R_{s,t+1})b_t$ log-linearizes around the steady state (b^*, s^*) as

$$r_{s,t+1} = \log(b_{t+1} + s_{t+1}) - \log b_t = k_0 + k_1 \log b_{t+1} + (k_2 \delta_b - 1) \log b_t,$$

where $k_1 = b^*/(b^* + s^*)$, $k_2 = s^*/(b^* + s^*)$, $k_1 + k_2 = 1$, and

$$k_0 = \log(b^* + s^*) - k_1 \log b^* - k_2 \log s^*. \quad (51)$$

The constant k_0 is a primitive: it depends only on b^* and s^* , and in particular is independent of ω .

A.2 Household, Firms, and the Phillips Curve

Household. The representative household maximizes

$$\max_{\{C_t, L_t\}} \mathbb{E}_0 \sum_{t \geq 0} \beta^t \left(\log C_t - \frac{L_t^{1+\psi}}{1+\psi} \right) \quad (52)$$

subject to the nominal budget constraint

$$P_t C_t + Q_t B_t^h + P_t a_t + P_t s_t = W_t L_t + I_t + B_{t-1}^h (1 - \omega) + \omega Q_t B_{t-1}^h + P_t a_{t-1} (1 + r_{t-1}), \quad (53)$$

where B_t^h is the household's nominal bond holding, a_t a real risk-free asset in zero net supply with gross real return r_t , W_t the nominal wage, I_t a lump-sum transfer of firm profits, and s_t the primary

surplus (a lump-sum tax). The Euler condition requires the log SDF m_{t+1} to satisfy

$$\log \mathbb{E}_t[\exp(m_{t+1} + r_{t+1}^x)] = 0 \quad (54)$$

for each excess log-return r_{t+1}^x , and the intratemporal condition yields the real-wage schedule $w_t = C_t L_t^\psi$. With log utility and goods-market clearing $C_t = Y_t$,

$$m_{t+1} - \mathbb{E}_t[m_{t+1}] = -(y_{t+1} - \mathbb{E}_t[y_{t+1}]). \quad (55)$$

Firms and the NKPC. Each firm i produces $Y_t(i) = A_t L_t(i)$ and faces CES demand $Y_t(i) = (P_t(i)/P_t)^{-\epsilon_p} Y_t$ with $\epsilon_p > 1$. A fraction $1 - \theta$ of firms, drawn at random each period, may reset. A resetting firm chooses P_t^* to maximize

$$\max_{P_t^*} \mathbb{E}_t \sum_{k=0}^{\infty} \theta^k M_{t,t+k} \frac{Y_{t+k|t}}{P_{t+k}} \left[P_t^* - MC_{t+k} \right], \quad Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\epsilon_p} Y_{t+k}, \quad (56)$$

with $M_{t,t+k} = \exp(\sum_{j=1}^k m_{t+j})$ the multi-period SDF and MC_t nominal marginal cost.

The Phillips curve with a risk wedge. Write the first-order condition in the recursive form (11). Under conditional log-normality each expectation contributes a Jensen term, so with $n_t \equiv \log \mathcal{N}_t$, $d_t \equiv \log \mathcal{D}_t$,

$$\begin{aligned} \hat{n}_t &= (1 - \beta\theta)(\hat{y}_t + \widehat{mc}_t) + \beta\theta \mathbb{E}_t[\hat{n}_{t+1} + \epsilon_p \pi_{t+1}] + \beta\theta J_t^N, \\ \hat{d}_t &= (1 - \beta\theta)\hat{y}_t + \beta\theta \mathbb{E}_t[\hat{d}_{t+1} + (\epsilon_p - 1)\pi_{t+1}] + \beta\theta J_t^{\mathcal{D}}, \end{aligned}$$

with $J_t^N = \frac{1}{2} \text{var}_t[m_{t+1} + \epsilon_p \pi_{t+1} + \hat{n}_{t+1}]$ and $J_t^{\mathcal{D}}$ defined analogously with $\epsilon_p - 1$ and $\hat{d}_{t+1}$. Subtracting, using $x_t \equiv \hat{n}_t - \hat{d}_t$ and the price index $\pi_t = \frac{1-\theta}{\theta} x_t$, gives (28),

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa \widehat{mc}_t + \Lambda_t, \quad \kappa = \frac{(1 - \theta)(1 - \beta\theta)}{\theta}, \quad \Lambda_t = \beta(1 - \theta)(J_t^N - J_t^{\mathcal{D}}).$$

Because the two recursions differ by exactly one power of gross inflation, the difference of their Jensen terms is, to leading order,

$$J_t^N - J_t^{\mathcal{D}} = \frac{2\epsilon_p - 1}{2} \text{var}_t(\pi_{t+1}) + \text{cov}_t(m_{t+1}, \pi_{t+1}) + O(\|\hat{n} - \hat{d}\|^2),$$

which is (29). At first order $\Lambda_t \equiv 0$ by certainty equivalence; it is retained here only because the same order of terms generates the second root of (24), and discarding it on one side of the model while keeping it on the other would be inconsistent. Substituting the equilibrium loadings, $\text{var}_t(\pi_{t+1}) = (B_2 \sigma)^2 + s^2 \xi_t$ and $\text{cov}_t(m_{t+1}, \pi_{t+1}) = -\lambda B_2 \sigma - \mu s \xi_t$ with $s \equiv B_4 v$, so that $\Lambda_t = \Lambda_0 + \Lambda_\xi \xi_t$

with

$$\Lambda_\xi = \beta(1 - \theta) \left[\frac{2\epsilon_p - 1}{2} s^2 - \mu s \right]. \quad (57)$$

θ is pinned by κ . The slope and the reset probability satisfy $\kappa = (1 - \theta)(1 - \beta\theta)/\theta$, i.e. $\beta\theta^2 - (1 + \beta + \kappa)\theta + 1 = 0$, whose root in $(0, 1)$ is

$$\theta = \frac{(1 + \beta + \kappa) - \sqrt{(1 + \beta + \kappa)^2 - 4\beta}}{2\beta}.$$

At $\kappa = 0.10$, $\beta = 0.99$ this gives $\theta = 0.733$, a mean price duration $1/(1 - \theta) = 3.7$ quarters. The demand elasticity ϵ_p enters only through Λ_ξ , so it is irrelevant at first order and matters only for the magnitude of the risk wedge. The real-rate shock ε_t enters as a supply shock shifting real marginal cost.

A.3 Duration, Passthrough, and the Inflation Recursion

Step 1: the bond-return innovation. From the definition (12), only the rolled-over fraction ω carries a capital gain, so

$$r_{g,t+1} - \mathbb{E}_t[r_{g,t+1}] \approx \omega(\log Q_{t+1} - \mathbb{E}_t[\log Q_{t+1}]).$$

An inflation surprise at $t + 1$ raises the nominal rate by ρ_π through the Taylor rule (16), and because inflation itself follows $\pi_{t+1} = \rho_\pi \pi_t + \dots$, the expected rate k periods later rises by ρ_π^{k+1} times the surprise. The geometric bond capitalizes this path with weights ω^k :

$$\log Q_{t+1} - \mathbb{E}_t[\log Q_{t+1}] = - \sum_{k \geq 0} \omega^k \rho_\pi^{k+1} (\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]) = - \frac{\rho_\pi}{1 - \omega \rho_\pi} (\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]). \quad (58)$$

Multiplying by ω gives the bond-return feedback (18),

$$r_{g,t+1} - \mathbb{E}_t[r_{g,t+1}] = -\gamma(\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]), \quad \gamma = \frac{\omega \rho_\pi}{1 - \omega \rho_\pi}. \quad (59)$$

At $\omega = 0$, $r_{g,t+1} = 1/Q_t \in \mathcal{F}_t$ and $\gamma = 0$: there is no capital-gain component.

Step 2: the passthrough factor. Log-linearizing the budget constraint (13) gives the inflation innovation as the gap between the two return innovations,

$$\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}] = (r_{g,t+1} - \mathbb{E}_t[r_{g,t+1}]) - (r_{s,t+1} - \mathbb{E}_t[r_{s,t+1}]). \quad (60)$$

Substituting (59) and solving,

$$\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}] = -\frac{1}{1+\gamma} (r_{s,t+1} - \mathbb{E}_t[r_{s,t+1}]) = -PT (r_{s,t+1} - \mathbb{E}_t[r_{s,t+1}]), \quad (61)$$

which is (19), with $PT = 1/(1+\gamma) = 1 - \omega\rho_\pi$.

Step 3: shock loadings B_2 and B_4 . With the affine bond value (21), the surplus-return innovation is

$$r_{s,t+1} - \mathbb{E}_t[r_{s,t+1}] = k_1 \left(A_1 \sigma \varepsilon_{t+1} + A_2 \nu \sqrt{\xi_t} \eta_{t+1} \right), \quad (62)$$

so (61) gives $\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}] = B_2 \sigma \varepsilon_{t+1} + B_4 \nu \sqrt{\xi_t} \eta_{t+1}$ with

$$B_2 = -k_1 A_1 PT > 0, \quad B_4 = -k_1 A_2 PT \leq 0. \quad (63)$$

Duration dampens both by PT ; at $\omega = 0$ they are the one-period Fisher coefficients $-k_1 A_1$ and $-k_1 A_2$.

Step 4: expected inflation from the bond Euler equation. Let $m_{t+1}^s \equiv m_{t+1} - \pi_{t+1}$ be the nominal SDF. The no-arbitrage condition $\log \mathbb{E}_t[\exp(m_{t+1}^s + \log r_{g,t+1})] = 0$ expands under conditional log-normality to

$$\mathbb{E}_t[r_{g,t+1}] = i_t - \frac{1}{2} \text{var}_t[r_{g,t+1}] - \text{cov}_t(m_{t+1}^s, r_{g,t+1}), \quad (64)$$

where i_t enters through $\mathbb{E}_t[m_{t+1}^s] = -i_t$. Using (59),

$$\text{var}_t[r_{g,t+1}] = \gamma^2 [(B_2 \sigma)^2 + (B_4 \nu)^2 \xi_t], \quad (65)$$

$$\text{cov}_t(m_{t+1}^s, r_{g,t+1}) = \gamma [(\lambda + B_2 \sigma) B_2 \sigma + (\mu + B_4 \nu) B_4 \nu \xi_t]. \quad (66)$$

Expected inflation then follows from the budget identity $\mathbb{E}_t[\pi_{t+1}] = \mathbb{E}_t[r_{g,t+1}] - \mathbb{E}_t[r_{s,t+1}]$.

Step 5: coefficient matching. Write $\mathbb{E}_t[\pi_{t+1}] = \rho_\pi \pi_t + \tilde{B}_0 + (B_1 + B_2 \rho_r) r_t + (B_3 + B_4 \rho_\xi) \xi_t$ and match against $\mathbb{E}_t[r_{g,t+1}] - \mathbb{E}_t[r_{s,t+1}]$. From (15) and condition (23), $\mathbb{E}_t[r_{s,t+1}] = \text{const} + r_t + A_2(k_1 \rho_\xi + k_2 \delta_b - 1) \xi_t$.

Coefficient of r_t . $\mathbb{E}_t[r_{g,t+1}]$ carries no direct r_t loading: the corrections (65)–(66) load only on ξ_t , and i_t loads on π_t . Hence

$$B_1 + B_2 \rho_r = -1 \quad (\text{Fisher consistency, for every } \omega), \quad (67)$$

so $B_1 = -1 - B_2 \rho_r = -1 + k_1 A_1 \rho_r PT < 0$.

Coefficient of ξ_t . Collecting the ξ_t terms gives (38),

$$B_3 + B_4\rho_\xi = A_2(1 - k_1\rho_\xi - k_2\delta_b) - \Xi(\omega), \quad \Xi(\omega) \equiv \gamma(\sigma_\pi^\eta + \mu)\sigma_\pi^\eta, \quad \sigma_\pi^\eta = B_4\nu. \quad (68)$$

At $\omega = 0$, $\gamma = 0$ and the Jensen correction Ξ vanishes, leaving the one-period FTPL fiscal channel $A_2(1 - k_1\rho_\xi - k_2\delta_b) > 0$.

Constant. Collecting the $\mathcal{F}_t$ -measurable terms in $\log \mathbb{E}_t[e^{m_{t+1}+r_{s,t+1}}] = 0$ reproduces (25), which solves for the intercept

$$A_0(\omega) = -\frac{k_0 + \frac{1}{2}(k_1A_1\sigma)^2 - \lambda(\omega)k_1A_1\sigma + k_1A_1(1 - \rho_r)\bar{r} + k_1A_2(\omega)(1 - \rho_\xi)\bar{\xi}}{k_1 + k_2\delta_b - 1}, \quad (69)$$

with $k_1 + k_2\delta_b - 1 < 0$ by active-fiscal stability. The inflation-recursion intercept is $B_0 = i^* - \rho_\pi\pi^* - k_0 - (k_1 + k_2\delta_b - 1)A_0(\omega)$. In the Fundamental Equilibrium ($A_2 = 0$) the intercept still shifts with ω through $\lambda(\omega)$; in the Fiscal Sunspot Equilibrium both $A_2(\omega)$ and $A_0(\omega)$ depend on ω .

A.4 Bond-Pricing Euler Equation

Substituting the SDF (22) and the conjecture (21) into the surplus return (15), the sum $m_{t+1} + r_{s,t+1}$ is conditionally Gaussian with shock loadings

$$c^\varepsilon = -\lambda + k_1A_1\sigma, \quad c^\eta = (-\mu + k_1A_2\nu)\sqrt{\xi_t}.$$

The Euler equation $\log \mathbb{E}_t[e^{m_{t+1}+r_{s,t+1}}] = 0$ therefore expands as

$$0 = \mathbb{E}_t[m_{t+1} + r_{s,t+1}] + \frac{1}{2}(-\lambda + k_1A_1\sigma)^2 + \frac{1}{2}(-\mu + k_1A_2\nu)^2\xi_t.$$

The Jensen correction $\delta = \frac{1}{2}(\lambda^2 + \mu^2\xi_t)$ inside m_{t+1} cancels the squared risk-price terms, leaving only cross terms. Collecting by r_t , ξ_t , and the constant yields conditions (23)–(25).

Coefficient of r_t . The drift of $k_1 \log b_{t+1}$ contributes $k_1A_1\rho_r r_t$, the $(k_2\delta_b - 1) \log b_t$ term contributes $(k_2\delta_b - 1)A_1r_t$, and $-r_t$ comes from the SDF. Setting the sum to zero gives (23), hence (26).

Coefficient of ξ_t . The drift of $k_1 \log b_{t+1}$ contributes $k_1A_2\rho_\xi \xi_t$, the $(k_2\delta_b - 1) \log b_t$ term contributes $(k_2\delta_b - 1)A_2\xi_t$, and the cross term from $\frac{1}{2}(-\mu + k_1A_2\nu)^2\xi_t$ after cancellation contributes $-\mu k_1A_2\nu + \frac{1}{2}k_1^2A_2^2\nu^2$. Factoring out A_2 gives (24). The root $A_2 = 0$ is the Fundamental Equilibrium. For the other, substitute μ from (32) with $B_4 = -k_1A_2PT$ and $B_3 + B_4\rho_\xi = A_2N - \Xi$. Setting the Jensen correction Ξ aside the bracket is linear in A_2 ,

$$-N + k_1^2A_2\nu^2\left[\frac{1}{2} + \frac{(1 - \beta\rho_\pi)PT}{\kappa} + \frac{\beta N}{k_1\kappa}\right] = 0,$$

which is the linear-in- A_2 case of (33); Appendix A.6 treats the general case.

The Jensen term and a spurious root. Restoring $\Xi = \gamma(\sigma_\pi^\eta + \mu)\sigma_\pi^\eta$ makes the bracket quadratic in A_2 , and since Ξ is itself linear in μ , the pair (μ, Ξ) solves in closed form at any candidate A_2 ; the resulting $\mu(A_2)$ carries the denominator $1 - (\nu\beta\gamma/\kappa)\sigma_\pi^\eta$. That denominator vanishes at

$$A_2^{\text{pole}} = \frac{\kappa}{\nu^2\beta\gamma PT k_1},$$

beyond which it changes sign and so does μ . The bracket then admits a further root at which $\mu > 0$ while $A_2 > 0$ — a bond simultaneously priced as a fiscal hazard and gaining real value with fiscal uncertainty, which is internally contradictory. This is an artifact of extrapolating a second-order correction into a region where it is no longer small: at the pole σ_π^η is roughly four times its value at the economic root. The solution reported throughout is the root below the pole, and with that restriction the model has exactly two equilibria.

A.5 Risk Prices

Take the unexpected component of the Phillips curve (28):

$$\kappa(y_{t+1} - \mathbb{E}_t[y_{t+1}]) = \Delta\pi_{t+1} - \beta(\mathbb{E}_{t+1}[\pi_{t+2}] - \mathbb{E}_t[\mathbb{E}_{t+1}[\pi_{t+2}]]) - \psi_r\sigma\varepsilon_{t+1}, \quad (70)$$

writing $\Delta\pi_{t+1} \equiv \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$. The middle term must be evaluated from the equilibrium recursion itself. Advancing (39) one period, $\mathbb{E}_{t+1}[\pi_{t+2}] = \rho_\pi\pi_{t+1} + C_0 - r_{t+1} + (B_3 + B_4\rho_\xi)\xi_{t+1}$, so its innovation carries *three* terms, not one:

$$\mathbb{E}_{t+1}[\pi_{t+2}] - \mathbb{E}_t[\mathbb{E}_{t+1}[\pi_{t+2}]] = \rho_\pi\Delta\pi_{t+1} - \sigma\varepsilon_{t+1} + (B_3 + B_4\rho_\xi)\nu\sqrt{\xi_t}\eta_{t+1}. \quad (71)$$

The second comes from $-r_{t+1}$ and the third from ξ_{t+1} ; both are first-order and neither may be dropped. Substituting into (70),

$$y_{t+1} - \mathbb{E}_t[y_{t+1}] = \frac{1 - \beta\rho_\pi}{\kappa}\Delta\pi_{t+1} + \frac{\beta - \psi_r}{\kappa}\sigma\varepsilon_{t+1} - \frac{\beta(B_3 + B_4\rho_\xi)\nu}{\kappa}\sqrt{\xi_t}\eta_{t+1}. \quad (72)$$

Combining with $m_{t+1} - \mathbb{E}_t[m_{t+1}] = -(y_{t+1} - \mathbb{E}_t[y_{t+1}])$, substituting the inflation innovation $\Delta\pi_{t+1} = B_2\sigma\varepsilon_{t+1} + B_4\nu\sqrt{\xi_t}\eta_{t+1}$ from (63), and matching against the SDF loadings $-\lambda$ and $-\mu\sqrt{\xi_t}$ yields (32):

$$\lambda = \frac{\sigma}{\kappa}[(1 - \beta\rho_\pi)B_2 + \beta - \psi_r], \quad \mu = \frac{\nu}{\kappa}[(1 - \beta\rho_\pi)B_4 - \beta(B_3 + B_4\rho_\xi)].$$

The two terms in λ have opposite signs, and $\lambda > 0$ at the baseline. Both terms in μ carry the sign of $-A_2$, since $B_4 \leq 0$ and $B_3 + B_4\rho_\xi \geq 0$, so $\mu < 0$ whenever $A_2 > 0$.

Remark on the ξ_{t+1} term. The $-\beta(B_3 + B_4\rho_\xi)v/\kappa$ loading in (72) is what couples the risk price back to the expected-inflation channel: a larger fiscal loading raises not only the impact inflation response but the whole expected path, and pricing that additional risk makes μ more negative. It is responsible for the third term in the coefficients of (33), and at the baseline it is the largest single contributor to $|\mu|$. Substituting $B_4 = -k_1 A_2 PT$ and $B_3 + B_4\rho_\xi = A_2 N - \Xi$ into the bracket condition of Appendix A.4 gives (33).

A.6 Reduction to a Quadratic and the Equilibrium Count

This appendix carries out the substitution behind Proposition 1. Write $a \equiv A_2$, $Q \equiv PT k_1$, $P \equiv Qv$, so that $B_4 = -Qa$ and the η -loading of inflation is $s = B_4 v = -Pa < 0$.

Step 1: the bond market gives μ linearly in a . Dividing the ξ_t condition (24) by $a \neq 0$ and solving,

$$\mu = c_1 a - c_0, \quad c_1 \equiv \frac{k_1 v}{2}, \quad c_0 \equiv \frac{N}{k_1 v} > 0. \quad (73)$$

Step 2: the goods market gives μ as a rational function of a . Substituting $B_3 + B_4\rho_\xi = aN - \gamma s(\mu + s) - \frac{1}{2}\gamma^2 s^2$ from (38) and Λ_ξ from (57) into (32), and collecting the terms in μ ,

$$\mu(1 + g a) = -h_1 a + h_2 a^2, \quad (74)$$

with

$$g = \frac{v\beta P}{\kappa}(\gamma + 1 - \theta), \quad h_1 = \frac{v}{\kappa}[(1 - \beta\rho_\pi)Q + \beta N], \quad h_2 = \frac{v\beta P^2}{\kappa} \left[\gamma \left(1 + \frac{\gamma}{2}\right) - \frac{(1-\theta)(2\epsilon_p - 1)}{2} \right].$$

All three are positive at the baseline except possibly h_2 , whose sign depends on ϵ_p .

Step 3: equating the two. Substituting (73) into (74) and rearranging gives the quadratic (33),

$$\underbrace{(c_1 g - h_2) a^2}_{\mathcal{A}} + \underbrace{(c_1 - c_0 g + h_1) a}_{\mathcal{B}} - c_0 = 0.$$

Factoring the common $\beta k_1^2 v^3 PT/\kappa$ out of $\mathcal{A}$ leaves the dimensionless expression (34) for S , and using $PT = 1/(1 + \gamma)$,

$$S = \frac{\gamma + 1 - \theta}{2} - \frac{\gamma}{1 + \gamma} - \frac{\gamma^2}{2(1 + \gamma)} + \frac{(1 - \theta)(2\epsilon_p - 1)}{2(1 + \gamma)}.$$

Step 4: counting roots. Since the constant term is $-c_0 < 0$ for any $N > 0$, the product of the roots is $-c_0/\mathcal{A}$ and their sum is $-\mathcal{B}/\mathcal{A}$. Therefore

- $\mathcal{A} > 0$ (i.e. $S > 0$): the roots have opposite signs, so exactly one is positive and the model has two equilibria;
- $\mathcal{A} < 0$ (i.e. $S < 0$): the product is positive and, since $\mathcal{B} > 0$ at the baseline, both roots are positive, giving three;
- $\mathcal{A} = 0$: the quadratic degenerates and the second root diverges.

Note that S depends only on $(\gamma, \theta, \epsilon_p)$ — not on $\nu, \bar{\xi}, N$ or any fiscal parameter. The equilibrium *count* is thus governed entirely by duration, price stickiness and the demand elasticity.

Step 5: why the large root is discarded. When $S < 0$ the second positive root lies at a roughly two orders of magnitude above the first. There the conditional standard deviation of inflation attributable to $\eta, |s|\sqrt{\xi_t} = PT k_1 a \nu \sqrt{\xi_t}$, is of the same order as inflation itself, so the second-order Jensen terms that generate the root are no longer small relative to the first-order terms they correct. The log-normal approximation on which (65)–(66) and (29) rest has broken down, and the root is an artifact of extrapolating that approximation outside its domain rather than a feature of the underlying economy. A useful diagnostic is the ratio of the Jensen correction to the FTPL term $A_2 N$ in (38): at the economic root it should be well below one, and at the large root it exceeds it. The solution reported throughout is the smallest positive root, and Section 5.5 reports the diagnostic at the baseline.

A.7 Welfare Criterion and the Maturity First-Order Condition

Second-order approximation to welfare. Take a second-order approximation to the household objective (3) around the efficient steady state. With log utility, Frisch parameter ψ , and Calvo price dispersion from (11), relative price dispersion satisfies $\text{var}_i \log P_t(i) = \theta \text{var}_i \log P_{t-1}(i) + \frac{\theta}{1-\theta} \pi_t^2$ to second order, so the discounted sum of dispersion is proportional to $\sum \beta^t \pi_t^2$ with factor $\epsilon_p \theta / [(1 - \theta)(1 - \beta\theta)] = \epsilon_p / \kappa$. Collecting terms gives (41),

$$\mathcal{W} \simeq -\frac{1}{2} \mathbb{E}_0 \sum_{t \geq 0} \beta^t \left[\frac{\epsilon_p}{\kappa} \pi_t^2 + (1 + \psi) \tilde{y}_t^2 \right] + \text{t.i.p.},$$

so the relative weight on the output gap is $\alpha = (1 + \psi)\kappa/\epsilon_p$ as in (42). This is the standard Woodford–Galí criterion; the only model-specific observation is that ϵ_p appears here as well as in the risk wedge (29), so a single parameter disciplines both.

Moments of the passthrough factor under the rule. Under (45), $PT_t = \overline{PT} - \rho_\pi \omega_\xi (\xi_t - \bar{\xi})$ with $\overline{PT} = 1 - \bar{\omega} \rho_\pi$. Since $\mathbb{E}[\xi_t - \bar{\xi}] = 0$ and $\mathbb{E}[(\xi_t - \bar{\xi})^2] = \text{Var}(\xi) = \nu^2 \bar{\xi} / (1 - \rho_\xi^2)$ exactly for the CIR

process,

$$\mathbb{E}[PT_t^2] = \overline{PT}^2 + \rho_\pi^2 \omega_\xi^2 \text{Var}(\xi), \quad \mathbb{E}[PT_t^2 \xi_t] = \overline{PT}^2 \bar{\xi} - 2\overline{PT} \rho_\pi \omega_\xi \text{Var}(\xi) + \rho_\pi^2 \omega_\xi^2 \mathbb{E}[(\xi_t - \bar{\xi})^2 \xi_t],$$

which are (46) once the last term is dropped as $O(\omega_\xi^2)$. Substituting into $\text{var}_t(\pi_{t+1}) = PT_t^2 k_1^2 (A_1^2 \sigma^2 + A_2^2 v^2 \xi_t)$ and taking unconditional expectations gives (47).

The first-order condition. Differentiating (47) with respect to ω_ξ ,

$$\frac{\partial}{\partial \omega_\xi} \mathbb{E}[\text{var}_t(\pi_{t+1})] = 2k_1^2 \rho_\pi \text{Var}(\xi) \left[A_1^2 \sigma^2 \rho_\pi \omega_\xi - A_2^2 v^2 \overline{PT} \right],$$

which is zero at (48), $\omega_\xi^* = A_2^2 v^2 \overline{PT} / (A_1^2 \sigma^2 \rho_\pi)$. The second derivative is $2k_1^2 \rho_\pi^2 \text{Var}(\xi) A_1^2 \sigma^2 > 0$, so this is a minimum of inflation volatility. Note that ω_ξ^* is proportional to A_2^2 and therefore vanishes in the Fundamental Equilibrium: with fiscal uncertainty unpriced there is nothing for state-contingent maturity to hedge.

The liquidity derivative. With $\ell_t = (1 - \omega_t)b_t$ and ω_t from (45), $\partial \ell_t / \partial \omega_\xi = -(\xi_t - \bar{\xi})b_t$, so

$$\frac{\partial}{\partial \omega_\xi} \mathbb{E}[v(\ell_t)] = \mathbb{E}[v'(\ell_t) \partial \ell_t / \partial \omega_\xi] = -\mathbb{E}[v'(\ell_t) b_t (\xi_t - \bar{\xi})] = -\text{Cov}(v'(\ell_t) b_t, \xi_t),$$

which is (49). For $v(\ell) = \ell^{1-\zeta} / (1 - \zeta)$ evaluated at $\omega_\xi = 0$, $v'(\ell_t) b_t = (1 - \bar{\omega})^{-\zeta} b_t^{1-\zeta}$, and since $\log b_t = A_0 + A_1 r_t + A_2 \xi_t$ with $A_2 > 0$ the covariance with ξ_t is positive for $\zeta < 1$, zero at $\zeta = 1$, negative for $\zeta > 1$. Combining with the inflation and output terms of (43) gives (50).

Why the constraint binds at the calibration. Evaluating (48): the sunspot numerator is $A_2^2 v^2 \overline{PT} = 3.53 \times 10^{-3}$ against a real-rate denominator $A_1^2 \sigma^2 \rho_\pi = 1.89 \times 10^{-3}$, a ratio of 1.9, so $\omega_\xi^* = 1.86$. Since $\text{sd}(\xi) = 0.64$, a one-standard-deviation move in fiscal uncertainty would call for $\Delta \omega = 1.19$, outside the admissible $[0, 1)$. The unconstrained optimum is therefore infeasible and the solution is at the upper corner; only the liquidity term in (50) pulls it back to an interior point. This is a statement about the calibration as much as the model — the ratio scales with $A_2^2 v^2$, which Section 4 flags as the least well identified block.

A.8 Backward-Looking Inflation and Forward-Looking Debt

The equilibrium inflation equation takes the form

$$\pi_{t+1} = \rho_\pi \pi_t + \underbrace{B_1 r_t + B_3 \xi_t}_{\text{backward channel}} + \underbrace{B_2 r_{t+1} + B_4 \xi_{t+1}}_{\text{FTPL forward channel}} + B_0. \quad (75)$$

This section explains why both sets of loadings are necessary, why the $\rho_\pi \pi_t$ term is unavoidable under passive monetary policy, and why the real debt value $\log b_t$ is by contrast purely forward-looking.

The budget constraint as the origin. The log-linearized nominal government budget constraint links tomorrow's real debt value, tomorrow's inflation, today's nominal rate, and today's surplus:

$$k_1 \pi_{t+1} = (k_1 - k_2 \delta_b) \log b_t + k_1 i_t - \log b_{t+1}. \quad (76)$$

Substituting the conjectures $\log b_t = A_0 + A_1 r_t + A_2 \xi_t$, $\log b_{t+1} = A_0 + A_1 r_{t+1} + A_2 \xi_{t+1}$, and the Taylor rule yields (75). Two channels emerge.

Channel 1 (backward): the nominal interest cost. The term $k_1 i_t$ is the nominal interest payment on yesterday's debt, serviced at $t + 1$. The central bank set this rate at date t via $i_t = \rho_\pi \pi_t + \text{const.}$ Because monetary policy is passive ($\rho_\pi < 1$), the nominal rate responds to inflation without raising the real rate one-for-one. This cost feeds through the budget identity to π_{t+1} , producing the backward terms $\rho_\pi \pi_t$, $B_1 r_t$, and $B_3 \xi_t$. The r_t and ξ_t loadings arise because π_t itself loads on the same state variables, and the AR(1) structure collapses all further history into a single lag. Even with $\delta_b = 0$ the interest-cost term would still create the backward link: the backward structure reflects *monetary policy setting the nominal rate on past information*, not the fiscal rule.

Channel 2 (forward): FTPL revaluation. The term $-\log b_{t+1}$ is the real value of bonds issued at $t + 1$, which the FTPL prices as the present value of future surpluses. A higher future real rate lowers the real debt value (since $A_1 < 0$), requiring higher inflation to equilibrate — hence $B_2 > 0$, damped by PT . Higher fiscal uncertainty raises the value loaded into bond prices when $A_2 > 0$, requiring lower inflation — hence $B_4 < 0$. These loadings depend on information realized *after* π_{t+1} is determined, entering through the no-arbitrage condition.

Why the inflation equation cannot load only on (r_{t+1}, ξ_{t+1}) . A solution $\pi_{t+1} = \Pi_0 + \Pi_r r_{t+1} + \Pi_\xi \xi_{t+1}$ with no lagged terms would require $B_1 = B_3 = 0$ and $\rho_\pi = 0$. Since $B_1 = -1 + k_1 A_1 \rho_r PT$, eliminating the lagged- r loading demands either $A_1 = 0$ (no real-rate loading at all) or a fiscal rule so aggressive it is inconsistent with active fiscal policy. More fundamentally, $\rho_\pi > 0$ is needed for the Taylor rule to anchor nominal expectations. A weaker condition eliminates only the *net* lagged effect: seeking a minimal-state-variable solution $\pi_t = C_0 + C_r r_t + C_\xi \xi_t$ and matching coefficients on r_t and r_{t-1} yields the knife-edge condition $\rho_\pi = k_1 - k_2 \delta_b$. At the baseline the right-hand side is ≈ 0.987 against $\rho_\pi = 0.85$, so the condition fails and the backward term is generic.

Why the bond value is purely forward-looking. The pricing equation $\log b_t = A_0 + A_1 r_t + A_2 \xi_t$ carries no dependence on r_{t-1} , ξ_{t-1} , or lagged debt. This is a consequence of *active fiscal policy* ($\delta_b < 1$): $\log b_t$ equals the present value of expected future surpluses, which depend on the path of future states, fully characterized by today's state through the Markov dynamics. There is no role for b_{t-1} because surpluses respond to debt only weakly, so the fiscal rule does not enforce repayment of accumulated past deficits. The real value of debt is determined by *asset pricing*, not backward accumulation. Under passive fiscal policy ($\delta_b > 1$, the Ricardian regime) the surplus retires any outstanding debt, b_t becomes a backward accumulation, and the FTPL channel breaks down.

A.9 Duration Comparative Statics

This appendix derives the closed-form partial derivatives behind Section 5.3. Every formula is verified against a central finite difference of the solver to a maximum relative error of 8.1×10^{-11} (src/scripts/make_omega_sensitivity.py).

All ω -dependence flows through one factor. The roll-over fraction enters the equilibrium *only* through

$$PT(\omega) = 1 - \omega \rho_\pi, \quad PT'(\omega) = -\rho_\pi < 0, \quad (77)$$

and through the way PT feeds the fiscal loading A_2^* . The objects k_1 , k_2 , and $A_1 = 1/(k_1 \rho_r + k_2 \delta_b - 1)$ are independent of ω , as is $N \equiv 1 - k_1 \rho_\xi - k_2 \delta_b > 0$.

Objects linear in PT . Everything that depends on ω only through PT , and does so linearly, has an exact closed-form derivative. That covers $B_2 = -k_1 A_1 PT$, $B_1 = -1 - B_2 \rho_r$, and — because λ depends on A_2 nowhere — the real-rate risk price:

$$\frac{\partial B_2}{\partial \omega} = -k_1 A_1 PT' = k_1 A_1 \rho_\pi < 0, \quad (78)$$

$$\frac{\partial B_1}{\partial \omega} = -\rho_r \frac{\partial B_2}{\partial \omega} > 0, \quad (79)$$

$$\frac{\partial \lambda}{\partial \omega} = \frac{\sigma(1 - \beta \rho_\pi)}{\kappa} \frac{\partial B_2}{\partial \omega} < 0. \quad (80)$$

Longer duration shrinks the fundamental real-rate passthrough B_2 and, via Fisher consistency (preserved exactly, $\partial(B_1 + B_2 \rho_r)/\partial \omega = 0$), shrinks $|B_1|$. These three are verified against central finite differences of the solver to a maximum relative error of 6.7×10^{-11} .

Objects routed through A_2^* . Once the Jensen correction Ξ is retained, A_2^* solves a quadratic rather than a linear condition, so A_2 , μ , B_4 and $B_3 + B_4 \rho_\xi$ no longer admit compact closed forms and

are differentiated numerically. Their signs are nonetheless unambiguous and stable across the parameter space:

$$\frac{\partial A_2^*}{\partial \omega} > 0, \quad \frac{\partial \mu}{\partial \omega} > 0, \quad \frac{\partial B_4}{\partial \omega} > 0. \quad (81)$$

Longer duration *raises* A_2^* : a lower passthrough shrinks the denominator coefficients of (33), so a larger debt-revaluation loading is self-consistent. At the same time $|\mu|$ *falls*, because μ is proportional to A_2^* times PT and the decline in PT dominates the rise in A_2^* . The two metrics therefore move in opposite directions with maturity, and statements about the “distance” between the equilibria have to name which one they use.

Expected inflation and the fiscal-sunspot loading. From (39), holding the states fixed, the economically relevant object is the cross-partial — how duration changes the slope of expected inflation in fiscal uncertainty:

$$\frac{\partial^2 \mathbb{E}_t[\pi_{t+1}]}{\partial \xi_t \partial \omega} = \frac{\partial(B_3 + B_4 \rho_\xi)}{\partial \omega} = N \frac{\partial A_2^*}{\partial \omega} - \frac{\partial \Xi}{\partial \omega} > 0, \quad (82)$$

with Ξ the Jensen correction of (68). Hence longer duration *amplifies* the passthrough of fiscal uncertainty to expected inflation, and since $\text{IRP}_t = -(B_3 + B_4 \rho_\xi) \xi_t$,

$$\frac{\partial \text{IRP}_t}{\partial \omega} = -\frac{\partial(B_3 + B_4 \rho_\xi)}{\partial \omega} \xi_t < 0 : \quad (83)$$

the safe-haven premium *deepens* with duration. The contrast with the fundamental channel (78) is exact and of opposite sign, which is the asymmetric maturity result of Section 5.3.

A.10 Decomposition of a Maturity (QE) Shock

This appendix derives the four-channel decomposition of Table 6. A QE shock is an MIT shock to the roll-over fraction, $\omega_H \rightarrow \omega_L$, reverting as $\omega_t = \omega_H + (\omega_L - \omega_H) \rho_\omega^t$. Since ω enters only through PT and through PT 's effect on A_2^* , every object can be re-solved at each ω_t and the impact response of expected inflation split by source.

Term-premium channels. From the expected bond return (64), write the term premium implicit in the covariance term as TP_t , so that the bond's expected excess return is $\gamma_t \text{TP}_t$. A change in maturity moves this product through two conventional channels,

$$\underbrace{\text{TP}_H \Delta \gamma_t}_{\text{(i) quantity}} + \underbrace{\gamma_t \Delta \text{TP}_t}_{\text{(ii) repricing}}, \quad (84)$$

which are the channels isolated by Corhay et al. [2023]. They are quantitatively small here and largely offsetting: as ω falls, fewer units of duration multiply the premium, while each unit's premium rises because a higher PT raises the inflation exposures σ_π^ε and σ_π^η .

Balance-sheet channels. The remaining response operates through the FTPL constant condition (69), in which ω enters through both $\lambda(\omega)$ and $A_2(\omega)$. Because λ is given by the same formula in both equilibria and does not involve A_2 , the λ -driven part of ΔA_0 is common to Eq. I and Eq. II. Writing $\Delta A_0^{(\lambda)}$ for the intercept change computed in the Fundamental Equilibrium (where $A_2 \equiv 0$), the impact inflation innovation splits as

$$\underbrace{-PT_t k_1 \Delta A_0^{(\lambda)}}_{\text{(iii-a) } A_0 \text{ hedge, both equilibria}} + \underbrace{-PT_t k_1 \left[(\Delta A_0 - \Delta A_0^{(\lambda)}) + \Delta A_2 \bar{\xi} \right]}_{\text{(iii-b) } A_2 \text{ coordination, Eq. II only}}. \quad (85)$$

Channel (iii-a) is the *balance-sheet hedge*: a lower ω raises PT , which makes $k_1 A_1 PT$ more negative in (32) and moves λ , repricing the debt intercept. It is present whether or not fiscal uncertainty is priced. Channel (iii-b) collects the two routes by which A_2 matters: the shift it induces in A_0 through the certainty-equivalent term $k_1 A_2 (1 - \rho_\xi) \bar{\xi}$ in (69), and the direct repricing of the ξ -loading $\Delta A_2 \bar{\xi}$ in $\log b_t$. Both vanish identically when $A_2 \equiv 0$.

Why the sign reverses. At the baseline, channel (iii-a) contributes -12.9 bp and channel (iii-b) $+26.4$ bp, with the term-premium channels together under 1.2 bp. In the Fundamental Equilibrium only (i), (ii) and (iii-a) operate, so the total is negative; in the Fiscal Sunspot Equilibrium (iii-b) dominates and the total is positive. The reversal therefore requires A_2^* to be large enough that (iii-b) exceeds (iii-a) in magnitude. Since $\partial A_2^* / \partial \omega > 0$ by (81), this holds only for sufficiently long pre-shock maturity, which is exactly the pattern in Figure 5. The computation is implemented in `src/inflme/qe.py`.

A.11 Nominal Term Structure

A d -period nominal zero-coupon bond pays one unit of currency at $t + d$. Its log price satisfies the no-arbitrage recursion

$$-\log q_t^d = -\log \mathbb{E}_t[\exp(m_{t+1} + \log q_{t+1}^{d-1} - \pi_{t+1})], \quad q_t^0 \equiv 1. \quad (86)$$

Conjecture and seed. Conjecture that for every $d \geq 1$,

$$-\log q_t^d = \Phi_d \pi_t + A_0^d + A_1^d r_t + A_2^d \xi_t, \quad \Phi_d = \sum_{k=1}^d \rho_\pi^k = \frac{\rho_\pi(1 - \rho_\pi^d)}{1 - \rho_\pi}. \quad (87)$$

The coefficient Φ_d accounts for the compounding of the Taylor rule: each additional period of indexation multiplies the nominal rate by ρ_π . The one-period bond satisfies $-\log q_t^1 = i_t = \rho_\pi \pi_t + (i^* - \rho_\pi \pi^*)$, so $A_1^1 = A_2^1 = 0$ and $\Phi_1 = \rho_\pi$. Although $A_1^1 = A_2^1 = 0$, the one-period price is *not* independent of the states: π_t itself loads on r_{t-1} and ξ_{t-1} . The $d = 1$ case is therefore an internal seed, and the first maturity with explicit loadings on the *current* state is $d = 2$.

Inductive step. Assume (87) holds for d . The exponent in (86) is

$$X_{t+1} \equiv m_{t+1} - (1 + \Phi_d) \pi_{t+1} - A_0^d - A_1^d r_{t+1} - A_2^d \xi_{t+1}, \quad (88)$$

which, substituting the state dynamics, the inflation recursion (36) and the SDF (22), is conditionally Gaussian with shock loadings

$$c_d^\varepsilon = -\sigma [\lambda + (1 + \Phi_d) B_2 + A_1^d], \quad c_d^\eta = -[\mu + (1 + \Phi_d) B_4 \nu + A_2^d \nu]. \quad (89)$$

Since $\log \mathbb{E}_t[\exp X_{t+1}] = f_t + \frac{1}{2}(c_d^\varepsilon)^2 + \frac{1}{2}(c_d^\eta)^2 \xi_t$ with f_t the $\mathcal{F}_t$ -measurable part, matching (87) at maturity $d + 1$ gives the recursions

$$\Phi_{d+1} = \rho_\pi (1 + \Phi_d), \quad (90)$$

$$A_1^{d+1} = -\Phi_d + \rho_r A_1^d, \quad (91)$$

$$A_2^{d+1} = (1 + \Phi_d)(B_3 + B_4 \rho_\xi) + \rho_\xi A_2^d - \mu \nu g_d - \frac{\nu^2}{2} g_d^2, \quad g_d \equiv (1 + \Phi_d) B_4 + A_2^d, \quad (92)$$

$$A_0^{d+1} = A_0^d + (1 + \Phi_d) C_0 + (1 - \rho_r) \bar{r} A_1^d + (1 - \rho_\xi) \bar{\xi} A_2^d - \sigma^2 \lambda c_d - \frac{\sigma^2}{2} c_d^2, \quad c_d \equiv (1 + \Phi_d) B_2 + A_1^d. \quad (93)$$

In (91) the cross term $1 + (1 + \Phi_d)(B_1 + B_2 \rho_r) = -\Phi_d$ collapses via Fisher consistency (67), so with $A_1^1 = 0$ the recursion gives $A_1^d = -\sum_{j=0}^{d-2} \rho_r^j \Phi_{d-1-j} < 0$ for all $d \geq 2$: longer nominal bonds are more sensitive to current real rates.

Yields. The continuously compounded d -period nominal yield is

$$y_t^d = -\frac{\log q_t^d}{d} = \frac{\Phi_d}{d} \pi_t + \frac{A_0^d}{d} + \frac{A_1^d}{d} r_t + \frac{A_2^d}{d} \xi_t. \quad (94)$$

The term $A_2^d \xi_t / d$ controls how fiscal uncertainty loads onto long yields: in the Fiscal Sunspot Equilibrium both A_2^d and $B_3 + B_4 \rho_\xi$ are positive, so fiscal stress raises the entire nominal yield curve, with the effect compounding at longer maturities through (92). In the Fundamental Equilibrium $A_2^d \equiv 0$ and the curve is driven by the real rate alone.